

Filtered Descent for the Physical Takeya Incidence Source

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Abstract

We construct a finite, source-hereditary descent for shaded Takeya incidence measures. A symmetric joint packet law keeps the forgotten translation fibre, all support marginals, and every later restriction in one probability space. Exact-support determinants, literal face deletion, root exchange, conditional insertion kernels, and a reduced history tree then reduce the global filtered estimate to one root-cross physical energy.

The analytic closure is geometric. The planar Córdoba estimate [2] gives the two-dimensional base. In dimension three the sticky Takeya theorem and its sticky/non-sticky reduction close the root gate. At the next handoff, directionwise sampling and an unbiased fibre lift reduce the remaining power-small source to the marked four-dimensional sticky estimate of [9]. Hence the scalar descent closes through dimension four; in dimensions $d \geq 5$ the analogous root-cross gate remains an input.

The Selmer–Cartan tower [1] is used only as structural provenance for this concrete filtration: its all-support comparison identifies the determinant successor and records coefficient, confluence, and root–Moore labels without turning them into physical multiplicity. It supplies no Takeya estimate. The hereditary typed statement has the further, independent requirement of a positive Hall/Radon–Nikodym capacity on correlated pair sources; this is not implied by the tower or by the scalar sticky input.

Standing inputs and scope. The physical source is a finite direction-separated family of shaded δ -tubes. All branches and deletions are measured in the single incidence law (21) and its symmetric packet extension (17). The initial hereditary shading density obeys $\lambda_{\text{in}}^{-1} = \delta^{-o(1)}$ for the subpower formulation; otherwise its reciprocal is retained explicitly. The analytic inputs are the hereditary planar Córdoba estimate [2], the three-dimensional sticky and sticky/non-sticky theorems [7, 10], and the uniform marked four-dimensional estimate [9]. These close (142) in dimensions two, three, and four, respectively.

For compatibility with [1], fix finite support, coefficient, confluence, and Cartan ceilings and a finite set of odd coefficient primes. The tower supplies the normalized all-support controller and its Moore–Reedy reconstruction. In particular, a q^2 line records supplied confluence provenance; it is not the physical pair multiplicity and is never used as an analytic estimate. The scalar proof below needs only the concrete support filtration, the exact-support/face decomposition, the hereditary source ledger, and proper-history recombination. The additional motivic labels are carried passively and then removed by the normalized physical augmentation.

1 The finite physical descent ledger

The descent is carried by a concrete finite weighted incidence system. A state records a direction support I , a retained insertion history χ , and the ordinary incidence atom. Coefficient, valuation, and Cartan labels may be attached when comparison with [1] is desired, but they do not alter

the underlying physical measure. Support deletion forgets slots, history reduction passes to a proper predecessor, and root projection retains the forgotten-direction fibre density.

Write $\text{Inc}_{Q,T_0}^{\text{FSC,rec}}$ for this finite support/history system at a rooted cell. The compatible full-motive decoration is only the base change

$$\boxed{\text{Inc}_{Q,T_0}^{\text{FSC,full}} := \text{Inc}_{Q,T_0}^{\text{FSC,rec}} \times_{\text{SCart}^{\text{rec}}} \text{SCart}^{\text{full}} \simeq \text{Inc}_{Q,T_0}^{\text{FSC,rec}} \times_{\text{Mot}_{\text{rec}, \leq 1}} \text{Mot}^{\text{full}}.} \quad (1)$$

Thus valuation depth belongs to the passive Mot^{full} -factor, whereas the physical lift count $\mathfrak{q}$ remains in the weighted middle fibre. No summation over a decoration fibre is allowed to create tube multiplicity.

Only four concrete properties of this ledger enter the scalar proof.

1. The exact-support term is the rooted transverse determinant, and deleting a support slot gives its literal proper face.
2. Insertion kernels compose by conditional probability; hence insertion order, confluence, restriction, and rerooting are compatible on one joint source.
3. Every retained restriction of packet mass α has one-point marginals bounded by $\alpha^{-1}\mu_{\text{inc}}$. We refer to this marginal statement below as R5.
4. All histories in one load layer are summed before taking the physical L^2/L^1 estimate. We refer to this global confluence estimate as R6.

The first three properties are proved directly in the next four sections. For the fourth, if γ ranges over retained history blocks and

$$\rho_\gamma := \frac{d(\pi_\gamma)_*\mu_\gamma}{d\mu_{\text{inc}}},$$

then the unmarked observable is

$$\mathcal{E}_{\text{inc}}(f) := \sum_\gamma \rho_\gamma \mathcal{E}_\gamma(f_\gamma). \quad (2)$$

The sum in (2) is not replaceable by one normalized chart. The analytic work below recombines all strict predecessor histories and reduces the remaining part of R6 to (142).

The Selmer–Cartan input is limited to structural provenance. Its normalized controller has one universal exact-support generator b_I^{univ} and one latching generator $\beta_I^{\text{univ}} = 2b_I^{\text{univ}}$. Their motivic images satisfy

$$\alpha_I^{\text{lat,Mot}} = 2\alpha_I^{\text{root,Mot}}. \quad (3)$$

The physical images are respectively $[\mathfrak{Z}_{Q,T_0}^{(d)}]$ and $2[\mathfrak{Z}_{Q,T_0}^{(d)}]$, as verified below. This common-source comparison records normalization and successor provenance; it neither identifies a torsion line with a real determinant line nor infers physical nonvanishing from a motivic class.

2 The fixed-root exact-support bridge

We now construct the first, source-compatible part of that realization. Fix a cube Q , a root tube T_0 meeting Q , and the ordinary conditional law

$$\mu_{Q,T_0}(U) = \Pr(U \mid U \text{ is an admissible side tube at } (Q, T_0)).$$

For the root-symmetric FSC source below, admissible means the full Q -star; the atom $U = T_0$ and all repeated side tuples have zero determinant and therefore do not alter the exact-support detector. Put $W = e_{T_0}^\perp$, $r = d - 1$,

$$I = \{0, 1, \dots, r\}, \quad I^\circ = I \setminus \{0\},$$

and $s(U) = P_W e_U$. Thus I is the unified tower's full direction support, including the root, whereas I° indexes the transverse slots. An ordered side tuple

$$\mathbf{U} = (U_1, \dots, U_r) \quad \text{has law} \quad \mu_{Q, T_0}^{\otimes r}(\mathbf{U}),$$

and its exact-support detector is

$$D_I(\mathbf{U}) = \det_W(s(U_1), \dots, s(U_r)), \quad |I| = d, \quad |I^\circ| = r. \quad (4)$$

The physical packet cochain is not a chosen graph or a selected skeleton. It is the orthogonal direct sum over all ordered tuples,

$$\mathfrak{Z}_{Q, T_0}^{(d)} = \bigoplus_{\mathbf{U}} \mu_{Q, T_0}^{\otimes r}(\mathbf{U})^{1/2} D_I(\mathbf{U}) z_{\mathbf{U}}, \quad (5)$$

where the $z_{\mathbf{U}}$ are unit generators of the tuple blocks. If

$$M_{Q, T_0} = \mathbb{E}_{U \sim \mu_{Q, T_0}}[s(U)s(U)^\top],$$

then the ordered Cauchy–Binet identity gives the exact norm formula

$$\|\mathfrak{Z}_{Q, T_0}^{(d)}\|^2 = \sum_{\mathbf{U}} \mu_{Q, T_0}^{\otimes r}(\mathbf{U}) |D_I(\mathbf{U})|^2 = r! \det M_{Q, T_0}. \quad (6)$$

This is an identity on the ordinary incidence source. No exceptional tuple has been selected and no change of marginal has occurred.

Common-controller comparison of the exact-support lines. For the i -th side entry let L_i denote its transverse rank-one slot. Root conditioning identifies the physical determinant line with the rooted physical coefficient line

$$\Lambda^r W \cong L_1 \otimes \dots \otimes L_r =: \mathcal{J}_{I^\circ}^{\text{phys}}. \quad (7)$$

It is not identified with the unified tower's torsion detector $\mathcal{D}_I^{\text{Mot}} = \mathcal{M}_N[I] \otimes_{\mathbf{Z}} \mathbf{Z}[I]$. Instead both receive maps from the integral controller line $\mathcal{C}_I^{\text{univ}} = \mathbf{Z}b_I^{\text{univ}}$. For $|I| = d \geq 3$, in the standing prime-power all-support range of the complete supportwise Moore–Reedy realization, the comparison is

$$\begin{aligned} \Phi_{\text{phys}}(b_I^{\text{univ}}) &= [\mathfrak{Z}_{Q, T_0}^{(d)}], & \Phi_{\text{Mot}}(b_I^{\text{univ}}) &= \alpha_I^{\text{root, Mot}}, \\ \Phi_{\text{phys}}(\beta_I^{\text{univ}}) &= 2[\mathfrak{Z}_{Q, T_0}^{(d)}], & \Phi_{\text{Mot}}(\beta_I^{\text{univ}}) &= \alpha_I^{\text{lat, Mot}} = 2\alpha_I^{\text{root, Mot}}, & \beta_I^{\text{univ}} &= 2b_I^{\text{univ}}. \end{aligned} \quad (8)$$

This records the normalization without requiring a nonexistent additive map from $\mathbf{Z}/N$ to $\mathbf{R}$.

The two first nontrivial ranks display the structural comparison concretely. In dimension 3, the top coefficient is

$$\alpha_1 \beta_2 - \beta_1 \alpha_2 = \det(s(U_1), s(U_2)).$$

In dimension 4, the top UT_5 latching coefficient is

$$\mu_a \kappa_{bc} - \mu_b \kappa_{ac} + \mu_c \kappa_{ab} = \det(s(U_1), s(U_2), s(U_3)).$$

The degree-two boundary has zero projection to the corresponding central top line, so these coefficients define the required closed exact-support classes. These are the first two instances of the all-support bar–cobar comparison constructed below. Formula (6) is the corresponding norm identity; it is not itself used to infer the higher-rank comparison.

Support deletion is literal. The tuple blocks in (5) are direct summands. Deleting a side tube U deletes exactly the blocks whose tuple contains U . The differential, every proper Hall-face map, and the detector all preserve the remaining direct sum. Consequently the unnormalized detector energy can only decrease, and after deletion it is again exactly $r! \det M'_{Q,T_0}$ for the restricted raw measure. At the coordinate level, deleting $i \in I^\circ$ replaces the top wedge by its $I^\circ \setminus \{i\}$ face; in the exact-support quotient every such proper face is killed. This proves all side-face compatibilities at a fixed root without a geometric covering argument. The remaining face $I \setminus \{0\}$ changes the chosen root and is not supplied by this direct sum argument.

The source ledger. Each one-point marginal of $\mu_{Q,T_0}^{\otimes r}$ is exactly μ_{Q,T_0} . If a later descent step conditions the tuple law on an event A , then for every side-tube set B

$$\Pr(U_i \in B \mid A) \leq \frac{\mu_{Q,T_0}(B)}{\Pr(A)}. \quad (9)$$

Thus the R5 loss is at most $\delta^{-o(1)}$ precisely when the retained event has ordinary tuple mass at least $\delta^{o(1)}$. Before such a conditioning, R5 holds with constant 1. Equation (9), rather than cohomological nonvanishing, is the complete fixed-root source ledger.

Root projection and the antecedent. Let $\pi_{T_0} = P_W$. For a shaded side tube U , do not immediately replace its projection by a binary planar shading. Retain the fibre density

$$g_U(z) = \int_{\pi_{T_0}^{-1}(z)} \mathbf{1}_{Y(U)}(x) d\mathcal{H}^1(x), \quad \int_W g_U(z) dz = |Y(U)|. \quad (10)$$

The second identity is Fubini, so projected mass is exact. Moreover, in an oriented orthonormal frame beginning with e_{T_0} ,

$$\det(e_{T_0}, e_{U_1}, \dots, e_{U_r}) = \det_W(P_W e_{U_1}, \dots, P_W e_{U_r}). \quad (11)$$

Hence the physical antecedent of the root class is the measure-valued packet

$$(\{(\pi_{T_0} U, g_U)\}_U, \mu_{Q,T_0}),$$

not merely the set of projected tube axes. Equations (10) and (11) prove compatibility with root projection while retaining both the forgotten-direction fibre and the ordinary one-point law. Converting g_U to a strong binary shading is a later physical estimate and must record the discarded mass.

The construction in this section proves the root-conditioned determinant identity, literal side-face compatibility, and the fixed-root marginal bound. The following sections extend the weighted source realization to insertion, confluence, the root face, rerooting, and global shaded restrictions. The extension to the all-support structural controller is given by the normalized bar-cobar construction below.

3 Insertion spans and confluence at one root

Fix (Q, T_0) , the full support I , and one joint ordinary law $\mathbf{P}_I$ on fully marked ordered side tuples. The product law used in (5) is one example, but independence is not needed here. For $A \subseteq I$ containing the root slot, let X_A be the finite set of A -marked physical packets and let $\mathbf{P}_A$ be the marginal of $\mathbf{P}_I$.

For $A \subseteq B$, define the insertion-history middle set

$$H_{A,B} := \{(x_A, x_B) \in X_A \times X_B : x_B|_A = x_A\}, \quad (12)$$

with its two projection legs. On the support of $\mathbf{P}_A$, give this span the conditional kernel

$$K_{A,B}(x_B \mid x_A) := \frac{\mathbf{P}_B(x_B)}{\mathbf{P}_A(x_A)} \mathbf{1}_{\{x_B|_A=x_A\}}. \quad (13)$$

Zero-mass states are discarded from the ordinary source. Then

$$\sum_{x_B|_A=x_A} K_{A,B}(x_B \mid x_A) = 1, \quad \mathbf{P}_B = K_{A,B} \mathbf{P}_A.$$

Thus every insertion span preserves the ordinary packet law exactly; R5 has constant 1 before any later conditioning.

For $A \subseteq B \subseteq C$, restriction gives a canonical identification

$$H_{A,B} \times_{X_B} H_{B,C} \cong H_{A,C},$$

and the kernels obey

$$K_{A,C}(x_C \mid x_A) = K_{B,C}(x_C \mid x_C|_B) K_{A,B}(x_C|_B \mid x_A). \quad (14)$$

This is the conditional-probability chain rule. In particular, for two new slots $i, j \notin A$, both insertion orders give the same weighted span:

$$K_{A \cup \{i\}, A \cup \{i,j\}} K_{A, A \cup \{i\}} = K_{A \cup \{j\}, A \cup \{i,j\}} K_{A, A \cup \{j\}} = K_{A, A \cup \{i,j\}}. \quad (15)$$

All longer confluence diagrams follow by associativity of (14).

The corresponding pull-push operator is

$$\mathbf{L}_{A,B} e_{x_A} = \sum_{x_B|_A=x_A} K_{A,B}(x_B \mid x_A) e_{x_B}. \quad (16)$$

Equations (14) and (15) say exactly that weighted pullback composition becomes matrix composition. Hence

$$\mathbf{L}_{B,C} \mathbf{L}_{A,B} = \mathbf{L}_{A,C}$$

strictly on the ordinary incidence module.

The raw lift count

$$\mathfrak{q}_{A,B}(x_A) := \#\{x_B \in X_B : x_B|_A = x_A\}$$

is the cardinality of the same middle fibre. It is retained by the span even though the conditional kernel has total mass 1. After restricting to an event E , the new conditional marginal is bounded by $\mathbf{P}(E)^{-1}$ times the old one, exactly as in (9).

This proves the insertion and confluence part of the finite physical realization. The endpoint-invisible A_N -frontier of the arithmetic FSC carrier is not identified with a $\mathfrak{q}_{A,B}$ -element physical set. It remains the Moore/root coefficient decoration, and its comparison with the physical exact-support line is the common-controller cospan (8). The following sections construct the root-face/rerooting span and compatibility of the rooted functors on cube overlaps.

4 The root-face span

Take a measurable partition by physical cells Q of diameter $O(\delta)$, and let

$$Y_Q(T) := Y(T) \cap Q, \quad w_{Q,T} := |Y_Q(T)|, \quad W_Q := \sum_T w_{Q,T}, \quad W := \sum_Q W_Q.$$

Discard zero-mass cubes and tube cells. On ordered d -tuples through one cell, allowing repeated entries, define

$$\mathbf{P}_w(Q, T_0, \dots, T_r) := \frac{W_Q}{W} \prod_{j=0}^r \frac{w_{Q, T_j}}{W_Q} = \frac{\prod_{j=0}^r w_{Q, T_j}}{W W_Q^r}, \quad r = d - 1. \quad (17)$$

Equivalently, sample the coarse shaded incidence (Q, T_0) with mass $w_{Q, T_0}/W$, and then sample r side tubes independently from the weighted Q -star with probabilities $w_{Q, U}/W_Q$. Repeated entries lie on proper diagonal faces and their determinant is zero.

The expression in (17) is invariant under every permutation of $(T_0, \dots, T_r)$. Moreover, every slot has exactly the coarse shaded-incidence marginal:

$$\sum_{T_0, \dots, \widehat{T_j}, \dots, T_r} \mathbf{P}_w(Q, T_0, \dots, T_r) = \frac{w_{Q, T_j}}{W}. \quad (18)$$

Conditional on (Q, T_0) , this is precisely the product law used in (5), with the weighted full Q -star as the side law.

For $j \in I^\circ$, let ρ_{0j} move T_j into the root slot and retain the relative order of the other entries. The doubly rooted middle object

$$\mathcal{R}_{0j} := \{(Q, T_0, \dots, T_r)\}$$

with its old-root and new-root legs is therefore a measure-preserving span. The inverse span is $\mathcal{R}_{j0}$, and permutation identities give the root-exchange groupoid relations strictly.

The determinant marking is compatible with this span. In the new order, with T_j first, one has

$$\det(e_{T_j}, e_{T_0}, \dots, \widehat{e_{T_j}}, \dots, e_{T_r}) = (-1)^j \det(e_{T_0}, \dots, e_{T_r}). \quad (19)$$

Applying (11) with root T_j shows that the new transverse determinant is the old exact-support determinant multiplied by the same orientation sign. Thus ρ_{0j} carries $\mathfrak{Z}_{Q, T_0}^{(d)}$ to the T_j -rooted packet cochain, with no norm or source loss.

Deleting the old root is now factored as

$$T_0\text{-rooted } I\text{-packet} \xrightarrow{\rho_{0j}} T_j\text{-rooted } I\text{-packet} \xrightarrow{\text{del}_0} T_j\text{-rooted } I \setminus \{0\}\text{-packet}. \quad (20)$$

To avoid choosing j , take the disjoint union of (20) over $j \in I^\circ$ with weight $1/r$. Equation (18) makes both endpoint laws ordinary. The full-support determinant maps to zero in the exact-support quotient of the proper face, while the remaining tuple is precisely the lower-support physical carrier. This is the missing root-face span.

If a branch condition retains only an event E of the symmetric tuple law, the same construction has Radon–Nikodym loss at most $\mathbf{P}_w(E)^{-1}$. Hence a retained mass $\delta^{o(1)}$ gives the declared $\delta^{-o(1)}$ R5 loss. No root-degree or point-degree factor survives.

The local FSC functor is therefore complete on one cube: side insertion, confluence, side deletion, root exchange, and the root face have compatible weighted spans. The remaining source-level interface is descent on overlaps of cubes and shaded restrictions.

5 Global incidence gluing and shaded restrictions

Retain the partition and weights from the root-face construction. The ordinary incidence space is

$$\Omega_{\text{inc}} := \{e = (Q, T, x) : x \in Y_Q(T)\},$$

with probability law

$$d\mu_{\text{inc}}(Q, T, x) := \frac{1}{W} \mathbf{1}_{Y_Q(T)}(x) dx, \quad m_a(x) := \sum_T a(Q(x), T, x) \mathbf{1}_{Y(T)}(x) \quad (21)$$

for every incidence weight $0 \leq a \leq 1$; here $Q(x)$ is the unique cell containing x . For the current support $I = \{0, \dots, r\}$, the global ordered packet law is (17); its coarse one-slot marginal is (18). To retain point-dependent hereditary restrictions, lift (17) to

$$\begin{aligned} d\tilde{\mathbf{P}}_w(Q, (T_0, x_0), \dots, (T_r, x_r)) \\ := \mathbf{P}_w(Q, T_0, \dots, T_r) \prod_{j=0}^r \frac{\mathbf{1}_{Y_Q(T_j)}(x_j) dx_j}{w_{Q, T_j}}. \end{aligned} \quad (22)$$

Zero-mass tube cells have already been discarded. Forgetting the points recovers $\mathbf{P}_w$, while every marked-incidence slot has marginal

$$d(\Pi_j)_* \tilde{\mathbf{P}}_w(Q, T, x) = \frac{1}{W} \mathbf{1}_{Y_Q(T)}(x) dx = d\mu_{\text{inc}}(Q, T, x). \quad (23)$$

The lifted law is symmetric in the pairs (T_j, x_j) . In particular m_1 is the ordinary shaded multiplicity and $W = \int m_1$.

Since Q remains part of every state, all support deletions, insertions, confluence maps, and root exchanges preserve the cube label. The global source-level functor is therefore the weighted direct sum

$$\Phi^{\text{rec}} = \bigoplus_Q \Phi_Q^{\text{rec}}. \quad (24)$$

Now let $0 \leq a(e) \leq 1$ be a hereditary shaded restriction. Apply the symmetric packet mask

$$A(Q, (T_0, x_0), \dots, (T_r, x_r)) := \prod_{j=0}^r a(Q, T_j, x_j)$$

to $\tilde{\mathbf{P}}_w$, and let $\alpha = \mathbb{E}_{\tilde{\mathbf{P}}_w} A$ be its retained packet mass. After normalization,

$$d\tilde{\mathbf{P}}_{w,A} = \alpha^{-1} A d\tilde{\mathbf{P}}_w.$$

For every slot j and every incidence set B ,

$$\tilde{\mathbf{P}}_{w,A}((Q, T_j, x_j) \in B) \leq \alpha^{-1} \mu_{\text{inc}}(B). \quad (25)$$

Thus a restriction with $\alpha \geq \delta^{o(1)}$ has precisely the required $\delta^{-o(1)}$ R5 domination. Because the mask is symmetric, rerooting remains measure preserving. Taking all support marginals from the single restricted joint law makes the conditional kernels satisfy (14), so insertion and confluence also survive the restriction.

Finally, root projection glues by ordinary pushforward. In a chart rooted at T_0 , let $g_{Q,U}$ be the e_{T_0} -fibre density of $Y(U) \cap Q$. Then

$$\sum_Q \int_{e_{T_0}^\perp} g_{Q,U}(z) dz = \sum_Q |Y(U) \cap Q| = |Y(U)|. \quad (26)$$

Overlapping projected cells contribute additively; none is selected or discarded. Equations (18), (25), and (26) close the global source and marginal part of the local FSC realization.

6 Structural provenance and passive decorations

The preceding construction already gives the physical support filtration. We now record, in one place, its compatibility with the all-support controller of [1]. Let X_A be the finite weighted packet space in slots $A \subseteq I$, and set

$$\mathcal{H}_A^{\text{phys}} := \ell^2(X_A) \otimes \Lambda^{|A|} \mathbf{R}^d, \quad \mathcal{H}_I^{\text{phys}} := \bigoplus_{A \subseteq I} \mathcal{H}_A^{\text{phys}}.$$

For $i \notin A$, use the conditional kernel of the preceding section to define

$$\begin{aligned} P_i(e_{x_A}) &:= \sum_{x_{A \cup \{i\}} | A = x_A} K_{A, A \cup \{i\}}(x_{A \cup \{i\}} | x_A)^{1/2} e_{x_{A \cup \{i\}}}, \\ D_i(e_{x_A} \otimes \omega) &:= \sum_{x_{A \cup \{i\}} | A = x_A} K_{A, A \cup \{i\}}(x_{A \cup \{i\}} | x_A)^{1/2} e_{x_{A \cup \{i\}}} \otimes (e_i(x_{A \cup \{i\}}) \wedge \omega). \end{aligned} \quad (27)$$

Both operators vanish after slot i has been inserted. The kernel chain rule (14) gives $P_i P_j = P_j P_i$; exterior creation therefore gives

$$D_i^2 = 0, \quad D_i D_j + D_j D_i = 0.$$

Conditioning on a root (Q, T_0) and contracting the top wedge with e_{T_0} yields

$$\det(P_{e_{T_0}^\perp} e_{U_1}, \dots, P_{e_{T_0}^\perp} e_{U_{d-1}}),$$

hence exactly the packet $\mathfrak{Z}_{Q, T_0}^{(d)}$.

The normalized cobar–Rees controller in [1] is generated by singleton insertions, with nonsingleton differentials equal to their signed commutators. Sending its singleton generator in slot i to P_i , and its exterior totalization to D_i , is therefore a dg representation:

$$\boxed{\Xi_I : \text{Tot}_\Lambda(\mathcal{U}_{I, N}^{\text{rec}}) \otimes_{\mathbf{Z}} \mathbf{R} \longrightarrow \text{End}(\mathcal{H}_I^{\text{phys}}).} \quad (28)$$

Its maximal bar word is the antisymmetrized product of the singleton insertions, so its normalized physical image is the top determinant just computed. This proves the common-controller comparison (8); deletion and permutation compatibility follow because both sides literally delete or relabel the same operators. No other part of the cobar or Moore presentation is used in the scalar estimate.

For completeness, the passive typed decorations can be audited without turning them into analytic hypotheses. At a bounded state $\mathcal{K}_{\eta; I, \chi, \nu}^{(n)}$, the confluence matching fibre is

$$M_{I, \chi}^{\text{conf}} := \text{holim}_{\chi' \prec \chi} \mathcal{K}_{\eta; I, \chi', \nu}^{(n)}, \quad F_{I, \chi}^{\text{conf}} := \text{fib} \left(\mathcal{K}_{\eta; I, \chi, \nu}^{(n)} \longrightarrow M_{I, \chi}^{\text{conf}} \right), \quad (29)$$

where $\chi' \prec \chi$ ranges over strict predecessor histories. The full structural audit is

$$\boxed{C_{I, \nu}^{\text{supp}}, \quad F_{\eta; I}^{\text{val}}, \quad F_{I, \nu}^{\text{depth}}, \quad F_{I, \chi}^{\text{conf}}, \quad \Theta^{\text{supp, val}}, \quad \Theta^{\text{supp, depth}}, \quad \text{Gr}_n^{\text{Cart}}.} \quad (30)$$

Only the exact-support determinant, the history/load fibre, and the physical root antecedent survive as scalar observables.

Indeed, over an incidence atom e , let $\mathcal{A}(e)$ be the finite fibre of compatible decorations and let $\kappa(a | e) \geq 0$ satisfy $\sum_{a \in \mathcal{A}(e)} \kappa(a | e) = 1$. Define

$$\epsilon_{\text{phys}} f(e) := \sum_{a \in \mathcal{A}(e)} \kappa(a | e) f(e, a). \quad (31)$$

Every transition in valuation, coefficient, or Cartan decoration fixes e and preserves this normalized kernel. It therefore acts as the identity after ϵ_{phys} ; support deletion commutes with it because the two maps act on separate coordinates. Thus vertical and mixed defects create no physical multiplicity. Confluence histories are different: they change the physical middle fibre and remain in the sum (2). In particular, the supplied q^2 provenance labels those histories but does not estimate their positive root-cross overlap.

7 Quantitative thickening of support latching

Exact support must be separated from a near-face packet at the physical scale. This is done inside the support filtration. Put $r = d - 1$ and, for an ordered rooted packet, define its Gram–Schmidt pivots by

$$p_1 := |s_1|, \quad p_j := \text{dist}(s_j, \text{span}(s_1, \dots, s_{j-1})) \quad (2 \leq j \leq r). \quad (32)$$

Then

$$|D_I(\mathbf{U})| = \prod_{j=1}^r p_j. \quad (33)$$

Fix a geometric threshold $\kappa \in [\delta, c_d]$ for a small dimensional constant c_d . The packet belongs to the κ -broad exact-support stratum when every $p_j \geq \kappa$. Otherwise assign it to the first failed pivot j , with lexicographic tie breaking. These $r + 1$ events are disjoint and exhaust the rooted packet law.

On the broad stratum,

$$|D_I(\mathbf{U})| \geq \kappa^r. \quad (34)$$

On the j -th near-face stratum let P_{j-1} be orthogonal projection to $\text{span}(s_1, \dots, s_{j-1})$ and put $\tilde{s}_j = P_{j-1}s_j$. Then

$$|s_j - \tilde{s}_j| < \kappa, \quad |D_I(s_1, \dots, s_r) - D_I(s_1, \dots, \tilde{s}_j, \dots, s_r)| \leq \kappa. \quad (35)$$

The second determinant on the left is zero. Thus the packet is a κ -thickening of the proper face $I \setminus \{j\}$, together with the linear-combination/history datum expressing $\tilde{s}_j$ in the earlier span.

This statement also has the required affine form. Write

$$v_j := \langle e_{U_j}, e_{T_0} \rangle e_{T_0} + \tilde{s}_j.$$

Then v_j belongs to the span of the root direction and the preceding side directions, while $|e_{U_j} - v_j| = |s_j - \tilde{s}_j| < \kappa$. For $\kappa < c_d$, $v_j \neq 0$ and $|e_{U_j} - v_j|/|v_j| = O(\kappa)$. Since every participating tube meets the current root cell, its unit-length segment lies in an $O(\kappa + \delta)$ -neighbourhood of the affine carrier generated by the root direction and the surviving pivot directions. Hence the failed pivot is realized by the proper-support stage with carrier width

$$a_{\text{face}} := \delta + \kappa. \quad (36)$$

The physical tube scale remains δ . The width a_{face} is a parameter of the physical realization, not valuation depth, coefficient depth, confluence multiplicity or obstruction height.

The carrier just constructed is initially a packet observable, not a cube observable. Give the broad stratum one formal mark c_{br} . If $\omega = (Q, T_0, U_1, \dots, U_r)$ lies instead in the j -th failed-pivot stratum, quantize the normal and affine offset of the span in the preceding paragraph and write

$$\mathbf{c}_{\text{pkt}}(\omega) := (j, \text{normal cap, offset bin}). \quad (37)$$

In general $\mathbf{c}_{\text{pkt}}$ depends on the completing tubes $U_1, \dots, U_r$; it is not determined by Q , nor by one terminal incidence (Q, T, x) . A cube label $\mathbf{c}_Q$ is available only after one proves a coherent

narrow-field statement: on the retained incidence set there is a unit normal ν_Q , depending only on Q , such that

$$|\nu_Q \cdot e_T| \lesssim \kappa \quad \text{for every retained } (Q, T). \quad (38)$$

Quantizing ν_Q and $\nu_Q \cdot x_Q$ at widths κ and a_{face} then gives $\mathfrak{c}_Q$. Every retained tube through Q lies in the corresponding fixed enlarged slab, and all terminal incidences at that cube have the same carrier label. Thus (38) is the precise passage from packetwise latching to a moving cube-carrier field; it is a physical source theorem, not a formal consequence of support deletion.

That passage can in fact be made on the intrinsic cube-narrow branch. With $\mu_Q(T) := w_{Q,T}/W_Q$, define the unrooted direction Gram operator

$$G_Q := \mathbb{E}_{T \sim \mu_Q} [e_T e_T^T].$$

The conditional form of the symmetric packet law (17) and Cauchy–Binet give

$$\mathbb{E}_{T_0, \dots, T_{d-1} \sim \mu_Q} |\det(e_{T_0}, \dots, e_{T_{d-1}})|^2 = d! \det G_Q. \quad (39)$$

Choose $\rho = \rho(\delta) \rightarrow 0$ with $\rho^{-1} = \delta^{-o(1)}$, and put $\beta_{\text{Gr}} := \rho^d \kappa^{2d}$. If $\det G_Q > \beta_{\text{Gr}}$, then, since the squared determinant is at most one, the packets satisfying

$$|\det(e_{T_0}, \dots, e_{T_{d-1}})|^2 \geq \frac{d!}{2} \beta_{\text{Gr}} \quad (40)$$

have conditional mass at least $c_d \beta_{\text{Gr}}$. This is a source-subpower broad packet event for subpower ρ^{-1} and κ^{-1} , and its determinant is exactly the physical image of the top exact-support word.

If $\det G_Q \leq \beta_{\text{Gr}}$, let ν_Q be a unit eigenvector for the smallest eigenvalue. Since $\text{tr } G_Q = 1$,

$$\mathbb{E}_{T \sim \mu_Q} |\nu_Q \cdot e_T|^2 = \lambda_{\min}(G_Q) \leq (\det G_Q)^{1/d} \leq \rho \kappa^2.$$

Markov's inequality deletes at most a ρ -fraction of the conditional incidence mass and leaves

$$|\nu_Q \cdot e_T| \leq \kappa. \quad (41)$$

This is (38) with a normal depending only on Q . Summing over cubes preserves the same deletion fraction. Thus the cube-first Gram split gives a source-compatible dichotomy: either retain a source-subpower broad packet population, or descend on a coherent moving cube-carrier field. The packetwise first-failed-pivot rule above remains the exact support-latching description, but it is not used to manufacture a cube label. After routing all packets in the intrinsic narrow branch to this valid cube carrier, the label is a function of the cube. The global coarse-carrier count below then controls both packet-completion marks and longitudinal fragmentation.

The first-failed-pivot partition is hereditary: after deleting tubes, shaded incidences or a support slot, recompute (32) on the restricted joint law. Every surviving atom still belongs to exactly one pivot stratum. Its conditional pushforward is therefore a normalized kernel, so R5 loses only the reciprocal mass of the retained stratum. Since r is fixed, either the broad stratum has the declared retained mass, or one proper-face stratum does, with only a dimensional loss.

Consequently the analytic support step has the precise dichotomy

broad exact support: $ D_I \geq \kappa^r$ and use the determinant/root projection; thickened proper face: $p_j < \kappa$ and descend to $I \setminus \{j\}$ in width $\delta + \kappa$.	(42)
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Choose $\kappa = \delta^{\varepsilon(\delta)}$, where $\varepsilon(\delta) \rightarrow 0$ and $\kappa \rightarrow 0$. Then the broad determinant floor κ^r is source-subpower. On the near-face branch the ordinary directional pair argument has only

$$L_{a_{\text{face}}} := 1 + \log_+(a_{\text{face}}/\delta) \leq 1 + \log(1/\delta)$$

in place of L ; it pays no power of the carrier width. Since support drops and d is fixed, the total number of such latching steps is finite and all threshold, chart and logarithmic losses remain $\delta^{-o(1)}$. This is the quantitative realization of support latching, not another motivic level.

8 From fibre densities to strong projected shadings

8.1 Binary hardening and the planar base

The measure-valued antecedent can be converted to a binary shading without choosing one lift. The conversion must be performed after aggregating all lifts in one projected line cell. Fix a root T_0 , write $\pi = \pi_{T_0} : \mathbf{R}^d \rightarrow W = e_{T_0}^\perp$, and work on a dyadic projected-speed shell

$$\sigma \leq |\pi e_U| < 2\sigma, \quad \delta \leq \sigma \leq 1. \quad (43)$$

Let C be an $O(\delta)$ projected line cell, let $\mathbb{U}_C$ be its physical lifts, and regularize $m \leq |\mathbb{U}_C| < 2m$. If $|Y(U)| \geq \lambda\delta^{d-1}$, put

$$n_C(x) := \sum_{U \in \mathbb{U}_C} \mathbf{1}_{Y(U)}(x), \quad g_C(z) := \int_{\pi^{-1}(z)} n_C(x) d\mathcal{H}^1(x). \quad (44)$$

Thus $g_C = \sum_{U \in \mathbb{U}_C} g_U$; no new measure has been introduced.

One root-parallel fibre meets a lift in length $O(\delta/\sigma)$, the projected segment has $(d-1)$ -dimensional volume $O(\sigma\delta^{d-2})$, and Fubini gives

$$0 \leq g_C \lesssim m \frac{\delta}{\sigma}, \quad \int_W g_C \gtrsim \lambda m \delta^{d-1}. \quad (45)$$

Choose a sufficiently small dimensional constant $c > 0$ and define

$$A_C := \left\{ z : g_C(z) \geq c \lambda m \frac{\delta}{\sigma} \right\}. \quad (46)$$

The complement of A_C carries at most a small fixed fraction of the second quantity in (45). The first quantity then yields

$$\boxed{|A_C| \gtrsim \lambda \sigma \delta^{d-2}, \quad g_C \gtrsim \lambda m \frac{\delta}{\sigma} \text{ on } A_C, \quad \int_{A_C} g_C \gtrsim \lambda m \delta^{d-1}.} \quad (47)$$

This is the required binary strong shading. The factors σ and σ^{-1} compensate exactly.

The associated fibre cutoff is equally source-compatible. Inside one projected line cell only the forgotten scalar direction parameter remains. For two full directions separated by angle θ , the intersection of their unit δ -tubes is $O_d(\delta^d/\theta)$. Direction separation and dyadic summation therefore give

$$\int_{\mathbf{R}^d} n_C(x)^2 dx \lesssim_d m \delta^{d-1} L, \quad L := 1 + \log(1/\delta), \quad (48)$$

uniformly in m and σ . To restart the three-dimensional argument from the plane, the projected translation load must be kept rather than discarded. Put $d = 3$, write $W = e_{T_0}^\perp \cong \mathbb{R}^2$, and on the occupancy/speed shell set

$$a_C := \frac{m_C \delta}{\sigma}. \quad (49)$$

If $\mathcal{C}_\theta$ consists of the projected cells with direction θ , direction separation upstairs gives

$$0 < a_C \lesssim 1, \quad \sup_\theta \sum_{C \in \mathcal{C}_\theta} a_C \lesssim 1. \quad (50)$$

Indeed $\sum_{C \in \mathcal{C}_\theta} m_C \lesssim \sigma/\delta$: after the projected direction and speed shell are fixed, only one δ -separated scalar direction parameter remains. The factor a_C is also forced by Fubini, since $a_C \sigma \delta = m_C \delta^2$, the original three-dimensional source mass.

The plane controls the weighted projected marginal. Define

$$W_{\text{pr}}(z) := \sum_C a_C \mathbf{1}_{A_C}(z). \quad (51)$$

Parallel translated cells have bounded overlap, and two cells whose projected directions make angle ϑ intersect in area $O(\delta^2/(\vartheta + \delta/\sigma))$. Summing first over translations by (50) and then over the δ/σ -separated projected directions gives the geometric planar estimate

$$\boxed{\int_W W_{\text{pr}}^2 \lesssim \frac{L_\sigma}{\lambda} \int_W W_{\text{pr}}, \quad L_\sigma := 1 + \log(\sigma/\delta).} \quad (52)$$

No Fourier estimate is used here. Hence, with $K_{\text{pr}} = L_\sigma/(\tau\lambda)$ and $H_{\text{pr}} := \{W_{\text{pr}} \leq K_{\text{pr}}\}$,

$$\int_{H_{\text{pr}}} W_{\text{pr}} \geq (1 - O(\tau)) \int_W W_{\text{pr}}, \quad \|W_{\text{pr}} \mathbf{1}_{H_{\text{pr}}}\|_\infty \leq K_{\text{pr}} = \delta^{-o(1)}. \quad (53)$$

An unweighted projected multiplicity bound cannot replace (53). Tile a fixed planar square, for each of $\asymp \delta^{-1}$ directions, by $\asymp \delta^{-1}$ parallel δ -tubes and give every cell weight $a_C = \delta$. Then (50) holds and $W_{\text{pr}} \simeq 1$, but $\sum_C \mathbf{1}_{A_C} \simeq \delta^{-1}$ everywhere. Any subsets B_C with unweighted overlap at most K satisfy

$$\sum_C a_C |B_C| \lesssim \delta K, \quad \sum_C a_C |A_C| \simeq 1. \quad (54)$$

Thus subpower retention would force $K \gtrsim \delta^{-1+o(1)}$. The earlier unweighted translation-stable target was therefore too strong; it erased the fibre coordinate and was false already over the planar base.

8.2 The fibre-aware three-dimensional descent

The correct three-dimensional descendant keeps that coordinate. Write $x = (z, t) \in W \times \mathbb{R}$ and put $\ell := \delta/\sigma$. Over its projected cell, each lift $U \in \mathbb{U}_C$ contributes an interval of length $O(\ell)$ centred on an affine height function $t = h_U(z)$. Thus, up to fixed enlargements,

$$n_C(z, t) = \sum_{U \in \mathbb{U}_C} \mathbf{1}_{\{|t - h_U(z)| \lesssim \ell\}}, \quad \rho_C(z, t) := \frac{n_C(z, t)}{a_C}. \quad (55)$$

On A_C , equations (45)–(47) say

$$c\lambda \lesssim \int_{\mathbb{R}} \rho_C(z, t) dt \lesssim 1. \quad (56)$$

In particular the actual lifted load is the weighted mixture

$$N_3(z, t) := \sum_C n_C(z, t) \mathbf{1}_{A_C}(z) = \sum_C a_C \rho_C(z, t) \mathbf{1}_{A_C}(z), \quad (57)$$

and therefore

$$c\lambda W_{\text{pr}}(z) \lesssim \int_{\mathbb{R}} N_3(z, t) dt \lesssim W_{\text{pr}}(z). \quad (58)$$

High unweighted projected overlap can therefore be harmless when the short fibre intervals are separated in t ; inequality (54) was detecting a projection artefact.

The exact stronger three-dimensional target is now fibre-aware. After the planar mask H_{pr} , one needs a single hereditary $H_3 \subseteq \pi^{-1}(H_{\text{pr}})$ such that

$$\boxed{\int_{H_3} N_3 \geq s_3 \int_{\pi^{-1}(H_{\text{pr}})} N_3, \quad \|N_3 \mathbf{1}_{H_3}\|_\infty \leq K_3, \quad s_3^{-1}, K_3 = \delta^{-o(1)}.} \quad (59)$$

The planar calculation proves the marginal part of this statement completely. The remaining assertion is an affine-fibre confluence packing estimate: short intervals centred on the functions h_U may not align with power-sized multiplicity on a source-subpower portion of their total mass. Pairwise alignment is controlled by (48); the unresolved sum is over repeatedly reused, source-correlated cycle cores. In the notation below, it is precisely the global packing requirement (169), now with the affine heights retained.

The usual broad/narrow split can be expressed entirely in this fibre-aware source. For three lifts put

$$J(U_1, U_2, U_3) := |\det(e_{U_1}, e_{U_2}, e_{U_3})|. \quad (60)$$

The projected directions, speeds and affine height slopes together determine J . On $J \geq \kappa$, the common intersection has volume $O(\delta^3/\kappa)$. After partitioning directions into κ -caps, the number of transverse cap triples is $\kappa^{-O(1)} = \delta^{-o(1)}$, and endpoint multilinear Kekeya [3, 12] controls the sum of the pointwise-broad descendants. If no such triple occurs at a high-load point, elementary linear algebra puts its active directions in an $O(\kappa)$ -neighbourhood of one plane.

The latter alternative is not yet a planar problem. A unit tube with normal direction component $O(\kappa)$ still moves through a κ -thick slab; collapsing that coordinate forgets the affine heights in (55). Partitioning into κ -slabs and dilating the normal coordinate by κ^{-1} produces the same three-dimensional affine-fibre problem at relative width

$$\delta' = \frac{\delta}{\kappa}, \quad (61)$$

with an inherited carrier and confluence history. Thus the narrow branch is a scale iteration of (59), not an application of the bare planar union theorem. To sum that iteration one needs a source-compatible packing bound for the reused narrow carriers; this is the same cycle-core issue, now organized by carrier depth.

This identifies what a proof “from dimension two” must add. Planar tube geometry gives (52)–(53); the full motivic filtration records the affine heights, their confluence histories and cycle provenance; the imported sticky theorem and sticky/non-sticky reduction supply the positive terminal packing in (177). Together these give (59) with subpower retained mass. A near-full hereditary weighted version would be maximal-function strength and is not supplied by these inputs.

8.3 The marked three-to-four-dimensional handoff

The next handoff shows exactly why the three-dimensional theorem cannot simply be quoted again in dimension four. Put now $d = 4$, so that $W = e_{T_0}^\perp \cong \mathbb{R}^3$, and retain the same dyadic occupancy/speed shell. The projected cells $A_C \subset W$ are translated three-dimensional tubes, while

$$W_3(z) := \sum_C a_C \mathbf{1}_{A_C}(z), \quad N_4(z, t) := \sum_C a_C \rho_C(z, t) \mathbf{1}_{A_C}(z) \quad (62)$$

obey

$$c\lambda W_3(z) \lesssim \int_{\mathbb{R}} N_4(z, t) dt \lesssim W_3(z). \quad (63)$$

Thus the projected datum is a weighted multi-translation family carrying the affine interval kernels ρ_C . The weights and parallel translations can nevertheless be coalesced by the ordinary three-dimensional sticky output.

For each projected direction θ , after a harmless fixed normalization put $A_\theta = \sum_{C \in \mathcal{C}_\theta} a_C \leq 1$ and independently choose at most one cell in that direction, choosing C with probability a_C and choosing none with probability $1 - A_\theta$. Every realization ω is direction separated. After

rescaling length σ to one, apply the three-dimensional sticky theorem and its sticky/non-sticky reduction to the shadings A_C in that realization. They give retained shadings $B_C^\omega \subseteq A_C$. Put

$$M_\omega^{\text{ret}} := \sum_{C \text{ chosen}} \mathbf{1}_{B_C^\omega}, \quad M_\omega := \sum_{C \text{ chosen}} \mathbf{1}_{A_C}.$$

With constants uniform in ω ,

$$\int_W M_\omega^{\text{ret}} \geq s_p \int_W M_\omega, \quad \|M_\omega^{\text{ret}}\|_\infty \leq K_p, \quad s_p^{-1}, K_p = \delta^{-o(1)}. \quad (64)$$

Define the coalesced fractional shading

$$w_C(z) := \Pr_\omega\{C \text{ is chosen, } z \in B_C^\omega\}, \quad \widetilde{W}_3(z) := \sum_C w_C(z).$$

Taking expectations in (64) gives

$$0 \leq w_C \leq a_C \mathbf{1}_{A_C}, \quad \int_W \widetilde{W}_3 \geq s_p \int_W W_3, \quad \|\widetilde{W}_3\|_\infty \leq K_p. \quad (65)$$

This is one coalesced fractional physical subsource, not a sum of local masks. Such Radon–Nikodym weights are native to the global source ledger; finite rational approximation changes only subpower constants. Thus ordinary three-dimensional sticky geometry completely closes the weighted multi-translation projected part.

What it does not see is the correlation among the fibre kernels. Couple the projected subsource to them by

$$N_4^{\text{pr}}(z, t) := \sum_C w_C(z) \rho_C(z, t), \quad c\lambda \widetilde{W}_3(z) \lesssim \int_{\mathbb{R}} N_4^{\text{pr}}(z, t) dt \lesssim \widetilde{W}_3(z). \quad (66)$$

One sufficient remaining target is a single hereditary physical subsource $0 \leq \widetilde{N}_4 \leq N_4^{\text{pr}}$, obtained by deleting complete affine interval atoms rather than by separately chosen point masks for the cells, and requires

$$\begin{aligned} \int_{W \times \mathbb{R}} \widetilde{N}_4 &\geq s_f \int_{W \times \mathbb{R}} N_4^{\text{pr}}, \quad s_f^{-1} = \delta^{-o(1)}, \\ \sup_{\substack{z \in W, I \subset \mathbb{R} \\ |I| \geq \ell}} \frac{\int_I \widetilde{N}_4(z, t) dt}{|I| + \ell} &\leq K_f, \quad K_f = \delta^{-o(1)}, \quad \ell = \delta/\sigma. \end{aligned} \quad (67)$$

This absolute capacity is unchanged by splitting a weight among parallel translates and asks only for the multiplicity needed in the four-dimensional conclusion; dividing additionally by $\widetilde{W}_3(z)$ would impose an unneeded stronger relative equidistribution. Since every summand in (55) is an interval of length comparable to ℓ , fixed enlargement of I gives, away from harmless interval endpoints,

$$\widetilde{N}_4(z, t) \lesssim \ell^{-1} \int_{|u-t| \lesssim \ell} \widetilde{N}_4(z, u) du \lesssim K_f. \quad (68)$$

Equations (65) and (67), followed by the usual dyadic choice of a projected mass shell if necessary, therefore give the four-dimensional analogue of (59), retaining at least a $c\lambda s_p s_f$ fraction of the shell mass. This is a marked-fibre interface, not a second application of the unmarked three-dimensional theorem. Indeed, concentrating all normalized fibre mass in one common interval of length ℓ leaves $\widetilde{W}_3$ unchanged but makes $N_4^{\text{pr}} \simeq \widetilde{W}_3/\ell$ there. Projection alone has lost exactly the datum that (67) measures.

For the actual scalar conclusion one can use a weaker quadratic form, which interfaces directly with the confluence filtration. Define the marked fibre collision energy

$$\mathcal{E}_{\text{fib}} := \int_{W \times \mathbb{R}} (N_4^{\text{pr}}(z, t))^2 dz dt. \quad (69)$$

The sufficient estimate is

$$\boxed{\mathcal{E}_{\text{fib}} \leq A_{\text{fib}} \int N_4^{\text{pr}}, \quad A_{\text{fib}} = \delta^{-o(1)}}. \quad (70)$$

Indeed, for the global deletion budget $\tau = \tau(\delta)$ put

$$H_{\text{fib}} := \{(z, t) : N_4^{\text{pr}}(z, t) \leq A_{\text{fib}} \tau^{-1}\}, \quad \widehat{N}_4 := N_4^{\text{pr}} \mathbf{1}_{H_{\text{fib}}}. \quad (71)$$

Then size-biased Markov truncation gives

$$\int_{H_{\text{fib}}^c} N_4^{\text{pr}} \leq \frac{\tau}{A_{\text{fib}}} \mathcal{E}_{\text{fib}} \leq \tau \int N_4^{\text{pr}}, \quad \|\widehat{N}_4\|_{\infty} \leq A_{\text{fib}} \tau^{-1} = \delta^{-o(1)}. \quad (72)$$

This is one global physical mask. It gives the same absolute interval inequality as (67), with $K_f = A_{\text{fib}} \tau^{-1}$, and gives the desired cutoff directly; unlike the atomwise Frostman route, it need not retain complete fibre atoms.

8.4 History recombination and the marked terminal source

Disintegrate the convexified source by its already existing reduced confluence histories,

$$N_4^{\text{pr}} = \sum_{\gamma} v_{\gamma}, \quad v_{\gamma} \geq 0,$$

using the same Radon–Nikodym-corrected joint law. The prefix-tree calculation (137)–(139) applies without change. Conditional only on the established proper-predecessor cutoff $B_{\text{fib}}^{\text{pred}}$, it gives

$$\sum_{a(\gamma, \gamma') \neq r_*} \int v_{\gamma} v_{\gamma'} \lesssim \delta^{-o(1)} B_{\text{fib}}^{\text{pred}} \int N_4^{\text{pr}}. \quad (73)$$

Consequently the only new quadratic term is

$$\boxed{\sum_{a(\gamma, \gamma') = r_*} \int v_{\gamma} v_{\gamma'} \lesssim \delta^{-o(1)} B_{\text{fib}}^{\text{pred}} \int N_4^{\text{pr}}}. \quad (74)$$

At this handoff the convexified three-dimensional cutoff (65), together with the already established strict-predecessor induction, gives $B_{\text{fib}}^{\text{pred}} = \delta^{-o(1)}$ on the retained shell. Hence (73) and (74) imply (70).

Thus full motivic descent really does remove every proper-history fibre collision, but it does not estimate the root-born correlated collision. Equation (74) is the three-to-four-dimensional specialization of (142); the structural q^2 confluence provenance labels its pair atoms but supplies no positive bound for their physical overlap.

There is still a nontrivial unconditional packing gain inside this root term. Regularize the marked lift atoms to comparable physical source mass, call the resulting weighted cardinality N_{fib} , and select disjoint dyadic carrier boxes $Q \times J \subset W \times \mathbb{R}$ at scale r , where Q has side r and J has length r . The affine heights are uniformly Lipschitz on the speed shell, so a marked unit

tube crosses such a box in longitudinal length $O(r)$. If $\mathbb{V}_{Q,J}$ is the set of atoms assigned to the box, put

$$I_r^{\text{fib}} := \sum_{Q \times J} |\mathbb{V}_{Q,J}|. \quad (75)$$

Exactly the incidence-source calculation in (170)–(174), now in four dimensions, gives

$$\boxed{\bar{\mathbf{b}}_r^{\text{fib}} \lesssim \min \left\{ 1, \frac{r I_r^{\text{fib}}}{\lambda N_{\text{fib}}} \right\}.} \quad (76)$$

Indeed the portion of one shaded δ -tube inside an r -box has volume $O(r\delta^3)$, whereas its current source mass is at least $\lambda\delta^3$. Fractional weights from (65) cancel on the two sides, so no inverse power of a_C is introduced.

Consequently all fibre-carrier shells satisfying

$$\sum_r \frac{r I_r^{\text{fib}}}{\lambda N_{\text{fib}}} = o(1) \quad (77)$$

can be deleted inside the existing global source budget. Failure of this packing can occur only on scales with

$$I_r^{\text{fib}} \gtrsim \frac{\lambda N_{\text{fib}}}{r}. \quad (78)$$

If saturation persists down a nested chain of boxes, the source has the same number of available longitudinal slots as an all-scale sticky family, but now the boxes retain the affine fibre mark. This is the terminal marked-sticky part of (74). The three-dimensional theorem controls its projection and therefore already accounts for (65); it does not control which projected atoms occupy the same intervals J . A regular tube–box bipartite reuse graph in which each tube meets $D \simeq \lambda/r$ selected boxes saturates (78), independently of its cycle-rank statistics. Thus neither unlabelled cycle rank nor the motivic history count proves the missing positive estimate.

The saturated marked source has a final geometric split. For four active lifts define

$$J_4(U_1, U_2, U_3, U_4) := |\det(e_{U_1}, e_{U_2}, e_{U_3}, e_{U_4})|. \quad (79)$$

On $J_4 \geq \kappa$, four tubes have common intersection $O(\delta^4/\kappa)$; after partitioning directions into κ -caps, endpoint multilinear Kakeya [3, 12] controls the whole broad contribution with loss $\kappa^{-O(1)} = \delta^{-o(1)}$. If no transverse quadruple occurs, elementary linear algebra places all active directions in an $O(\kappa)$ -neighbourhood of one three-dimensional hyperplane. Partition into κ -slabs and dilate the normal coordinate by κ^{-1} . The relative width becomes

$$\delta' = \frac{\delta}{\kappa}, \quad (80)$$

and the affine interval marks, confluence histories and saturated carrier chain are all retained. Thus this narrow branch is the same four-dimensional marked-fibre problem at a coarser scale; it is not an ordinary three-dimensional source.

After the broad terms and the non-saturated shells have been removed, the exact terminal estimate is

$$\boxed{|U(\mathbb{V}, Y; \mathcal{Q}_{\text{fib}})| \geq \delta^{o(1)} \sum_{V \in \mathbb{V}} |Y(V)|, \quad \mu(\mathbb{V}, Y; \mathcal{Q}_{\text{fib}}) \leq \delta^{-o(1)}.} \quad (81)$$

Here $\mathcal{Q}_{\text{fib}}$ is the inherited nested carrier chain satisfying (78); it is a mark on the same physical union, not a second measure. The low-multiplicity truncation used after (177) closes the four-dimensional scalar refinement directly, bypassing the need to establish (74) on the unpruned source. The uniform marked source-hereditary theorem of [9] applies with weights w_V , arbitrary

fractional restrictions $0 \leq \tilde{w}_V \leq w_V$, the affine fibre as a conditional mark, and the δ -dependent tree $\mathcal{Q}_{\text{fib}}$. Its weighted L^2 estimate gives the volume bound, and its half-source-mass low-multiplicity mask gives the second conclusion in (81) after the permitted relabelling of the retained shading. Hence the four-dimensional scalar refinement is closed.

The strength of this input is essential. The earlier four-dimensional dimension-13/4 estimate [8] permits a power loss of order $\delta^{3/4+o(1)}$ in union volume, rather than the $\delta^{o(1)}$ retention required in (81). It therefore cannot be inserted into the subpower descent ledger. More generally, if the terminal input has only

$$|U(\mathbb{V}, Y; \mathcal{Q}_{\text{fib}})| \geq \delta^{\eta_4+o(1)} \sum_{V \in \mathbb{V}} |Y(V)|, \quad (82)$$

then the same low-average-multiplicity truncation gives only

$$s_4 \geq \delta^{\eta_4+o(1)}, \quad K_4 \leq \delta^{-\eta_4-o(1)}. \quad (83)$$

At the level of finite-scale exponent bookkeeping, the known 13/4 dimension corresponds to $\eta_4 = 3/4$; the cited Hausdorff-dimension statement is not being asserted here in the stronger uniform hereditary form (82). In either form the filtration preserves a positive exponent honestly; it does not convert a power loss into a subpower one.

One can now test the proposed lower-dimensional recursion quantitatively. Let $\mathfrak{R}_4(\delta)$ denote the worst normalized root-fibre collision ratio

$$\frac{\mathcal{E}_{\text{fib}}}{B_{\text{fib}}^{\text{pred}} \int N_4^{\text{pr}}}$$

over the retained terminal marked sources at width δ . The four-linear broad estimate and one narrow rescaling give, with the presently available carrier coalescing, at best the following schematic but loss-explicit inequality:

$$\mathfrak{R}_4(\delta) \lesssim \delta^{-o(1)} + \kappa^{-C} \mathfrak{R}_4(\delta/\kappa). \quad (84)$$

The polynomial factor is not a count of motivic histories. It is the physical congestion of the possible narrow hyperplane carriers and their affine fibre translations. Replacing it by the number of structural operation words would again confuse a geometric scale label with a confluence level.

Along a purely narrow chain choose scales $\delta_{j+1} = \delta_j/\kappa_j$ until $\delta_L \simeq 1$. Necessarily

$$\prod_{j=0}^{L-1} \kappa_j \simeq \delta, \quad \prod_{j=0}^{L-1} \kappa_j^{-C} \simeq \delta^{-C}. \quad (85)$$

Thus iteration of (84) has a power loss even when every three-dimensional projected source is closed perfectly. Taking $\kappa_j = \delta_j^\varepsilon$ merely distributes that same power among more scales; it does not make the product subpower.

A successful $3 \rightarrow 4$ recursion would therefore have to replace the narrow term by

$$\mathfrak{R}_4(\delta) \lesssim \delta^{-o(1)} + G(\kappa) \mathfrak{R}_4(\delta/\kappa), \quad \prod_j G(\kappa_j) = \delta^{-o(1)}. \quad (86)$$

The required improvement from κ^{-C} to such a scale-summable coefficient is exactly a coherent packing estimate for the affine fibre carriers. In the present notation it is (81), or on the quadratic route (74). For the common-interval example after (67), the fibre length changes from ℓ to ℓ/κ under rescaling, so its density obeys

$$\ell^{-1} = \kappa^{-1}(\ell/\kappa)^{-1}.$$

Thus this example alone forces a factor at least κ^{-1} in any recursion based only on the projected load. It rules out a scale-summable coefficient without an additional fibre-spreading estimate.

This also explains the fixed exponent in the known planebrush induction. Its trilinear estimate already contains a $\delta^{3/4}$ union-volume factor, while its plany branch is reassembled across scales using stickiness [8]. The induction propagates the exponent $\eta_4 = 3/4$; it does not define a map $\eta_4 \mapsto c\eta_4$ with $c < 1$ that could be iterated to zero. Hence the existing two- and three-dimensional theorems plus the published four-dimensional planebrush recursion do not prove full (81).

8.5 Randomized fibre lifting and occupancy

There is a sharper direct test of whether the three-dimensional convexification can be lifted without the missing terminal input. Restrict to a dyadic shell

$$a \leq a_C < 2a. \quad (87)$$

In the sampling construction preceding (64), define the unbiased lifted random load

$$Z_\omega(z, t) := \sum_{C \text{ chosen}} a_C^{-1} n_C(z, t) \mathbf{1}_{B_C^\omega}(z). \quad (88)$$

The factor a_C^{-1} is forced: taking expectation gives exactly the convexified physical subsource,

$$\mathbb{E}_\omega Z_\omega = N_4^{\text{pr}}. \quad (89)$$

Without this factor the expectation retains only an a_C -fraction of the original cell source and is unusable on a power-small occupancy shell.

For each realization the retained projected multiplicity is at most K_p . Hence Cauchy–Schwarz over its active projected cells, followed by (48), gives

$$\int Z_\omega^2 \lesssim K_p a^{-2} \sum_{C \text{ chosen}} \int n_C^2 \lesssim K_p L a^{-2} \sum_{C \text{ chosen}} m_C \delta^3.$$

In dimension four, $m_C \delta^3 = a_C \sigma \delta^2$; (47) gives $|A_C| \gtrsim \lambda \sigma \delta^2$ and $g_C \gtrsim \lambda a_C$ on A_C ; and (64) gives $\sum_C |B_C^\omega| \geq s_p \sum_C |A_C|$. Combining these relations yields

$$\boxed{\int Z_\omega^2 \lesssim \frac{K_p L}{a \lambda^2 s_p} \int Z_\omega.} \quad (90)$$

Jensen’s inequality and (89) therefore imply

$$\mathcal{E}_{\text{fib}} \lesssim \frac{K_p L}{a \lambda^2 s_p} \int N_4^{\text{pr}}. \quad (91)$$

Consequently every shell satisfying

$$a^{-1} = \delta^{-o(1)} \quad (92)$$

obeys (70) using only the completed three-dimensional sticky theorem, the geometric fibre energy and the global source convexification.

This argument also identifies why the remaining low-occupancy shell is not a bookkeeping artefact. Removing a_C^{-1} from (88) loses an a_C -fraction of the physical source; keeping it produces exactly the factor a^{-1} in (90). Selecting each cell more often creates a^{-1} parallel-translation color classes and returns the same congestion after coalescing. Thus the direct $2/3 \rightarrow 4$ recursion is now proved on the high-occupancy source, and its unresolved part is reduced further to

$$a = \delta^{\beta+o(1)} \quad \text{for some fixed } \beta > 0, \quad (93)$$

with correlated affine fibre intervals. This is precisely the source on which a new fibre-spreading gain must recover the factor a^{-1} .

8.6 The low-occupancy broad/narrow branch

The low-occupancy source admits a source-compatible pointwise broad/narrow split which sharpens this description. Let Γ_x be the normalized physical direction law of the active marked lift atoms at $x = (z, t)$, with all Radon–Nikodym weights retained, and put

$$G_x^{\text{fib}} := \int e_U e_U^\top d\Gamma_x(U). \quad (94)$$

Fix subpower parameters θ, κ and split the ordinary incidence mass according to

$$E_{\text{br}} := \{x : \det G_x^{\text{fib}} \geq \theta\}, \quad E_{\text{nar}} := \{x : \det G_x^{\text{fib}} < \theta\}. \quad (95)$$

If E_{br} carries a source-subpower fraction, pointwise Cauchy–Binet gives a source-subpower fraction of transverse four-tuples at the same physical point. After the usual cap regularization, endpoint multilinear Kakeya closes that descendant. This avoids the false implication from the averaged cube Gram operator noted in (186): broadness is tested directly on the common-point source.

On E_{nar} , choose measurably a unit eigenvector $n(x)$ for the least eigenvalue. Since $\text{tr } G_x^{\text{fib}} = 1$,

$$\lambda_{\min}(G_x^{\text{fib}}) \leq (\det G_x^{\text{fib}})^{1/4} < \theta^{1/4}.$$

Consequently Markov’s inequality in the conditional direction law gives

$$\Gamma_x\{U : |\langle e_U, n(x) \rangle| > \kappa\} \leq \frac{\theta^{1/4}}{\kappa^2}. \quad (96)$$

Choose $\theta^{1/4} \leq \tau\kappa^2$ with $\tau^{-1}, \kappa^{-1} = \delta^{-o(1)}$ and delete this source-small part. Quantize $n(x)$ by a κ -net in S^3 , of cardinality $O(\kappa^{-3}) = \delta^{-o(1)}$, and retain one normal bin of maximal ordinary incidence mass. This is one global source selection. On the resulting subpower descendant there is a fixed unit normal n_ν such that

$$|\langle e_U, n_\nu \rangle| \lesssim \kappa \quad \text{for every retained active incidence atom.} \quad (97)$$

Thus the pointwise-narrow part really does enter one coherent three-dimensional hyperplane carrier before rescaling; no point-dependent family of local masks is being summed.

However, applying (80) to (97) returns the same marked-fibre problem at width δ/κ . If its new occupancy is subpower it closes by (92); otherwise it remains in (93). Repeating the latter alternative gives precisely (84). Hence all low-occupancy mass that is pointwise broad at some scale closes; the only surviving configuration is an all-scale chain of coherent pointwise-narrow hyperplane carriers whose affine fibre density pays the unavoidable factor in (85). This is a smaller and fully physical form of the terminal marked-sticky obstruction.

Pulling the successive narrow normals back to the original coordinates gives a useful flag dichotomy. After rotating (97), write $n_\nu = e_4$ and let

$$A_\kappa := \text{diag}(1, 1, 1, \kappa^{-1})$$

be the narrow dilation. If $n' = v + \alpha e_4$ is the next narrow normal in the dilated coordinates, its pullback constraint on an original direction is

$$|\langle A_\kappa^\top n', e_U \rangle| \lesssim \kappa'. \quad (98)$$

If $|v| \gtrsim \kappa'$, this covector is quantitatively independent of e_4 ; the retained directions lie near the intersection of two hyperplanes. This is a proper two-dimensional direction plane and passes to the already closed lower-support branch. Such transverse flag drops can occur only finitely many times.

If $|v| \lesssim \kappa'$, then (98) does not introduce a new flag direction; instead it improves the original normal aperture from κ to

$$|\langle e_U, e_4 \rangle| \lesssim \kappa \kappa'. \quad (99)$$

Hence a purely coherent narrow chain eventually places every direction within $O(\delta)$ of one fixed hyperplane. That terminal flag is genuinely three-dimensional. Partition physical space into normal δ -slabs; each retained tube meets $O(1)$ adjacent slabs, and inside one slab its projection to $e_4^\perp$ has only bounded direction multiplicity. Fubini strong-shading selection followed by the three-dimensional sticky theorem therefore gives

$$|U(\mathbb{V}, Y)| \geq \delta^{o(1)} \sum_{V \in \mathbb{V}} |Y(V)| \quad (100)$$

on this terminal fixed-hyperplane descendant.

The obstruction lies between (97) and (100). Under A_κ , a round four-dimensional δ -tube becomes an anisotropic plank with transverse widths comparable to

$$(\delta, \delta, \delta/\kappa). \quad (101)$$

Enlarging the two thinner transverse axes to δ/κ in order to invoke an isotropic theorem pays a polynomial power of κ^{-1} at every coherent scale. This is the physical factor in (84); it is not caused by choosing new normals, since the coherent branch uses the same pulled-back flag.

Thus a lower-dimensional proof would close if one could establish the following eccentricity-uniform bridge for the inherited coherent flag:

$$|U(\mathbb{P}, Y; \mathcal{F})| \geq \delta^{o(1)} \sum_{P \in \mathbb{P}} |Y(P)|, \quad \text{uniformly in all intermediate plank aspect ratios.} \quad (102)$$

Here $\mathcal{F}$ is the single pulled-back normal flag, not a new descent level. The ordinary three-dimensional sticky theorem proves the terminal aspect ratio (100), but it does not provide uniform control while the normal width lies strictly between δ and one. The flag-anisotropic gate is therefore a still sharper formulation of the missing fibre-spreading estimate: proving it would make the coherent aperture products telescope to the three-dimensional base without the power loss in (85).

8.7 The flag tensorization obstruction

The anisotropic gate has a lower-dimensional tensorization interpretation. In coordinates adapted to the fixed flag, a coherent plank atom is determined by a tangential three-dimensional tube T_c and a two-dimensional strip S_j in the longitudinal–normal plane, with the same longitudinal parameter. Up to fixed enlargements,

$$P_{c,j} \simeq T_c \times_{[0,1]} S_j. \quad (103)$$

The three-dimensional sticky theorem controls the T_c marginal and the geometric planar Córdoba estimate controls the S_j marginal. Neither marginal estimate controls a physical selector whose two outputs depend jointly on (c, j) .

There is nevertheless a precise sufficient tensorization. Suppose the retained flag source has a positive fractional decomposition into modes M_η , and on every mode its physical kernel is endpointwise dominated by

$$K_\eta^T(h \mid c) K_\eta^S(h' \mid j).$$

If the two unary kernels have inherited RN constants $A_{\eta,T}$ and $A_{\eta,S}$, require

$$\sum_{\eta} t_\eta A_{\eta,T} A_{\eta,S} \leq \delta^{-o(1)}, \quad \sum_{\eta: (c,j) \in M_\eta} t_\eta \geq 1 \quad \text{on the positive flag source.} \quad (104)$$

Thus “mode” means an actual endpointwise factorizing positive source, not an arbitrary rectangle containing nonexistent pairs. The external-product calculation (201) applies on each mode; positive coalescence proves the RN/energy form of (102). This is the flag realization of the fractional criterion (204).

The general low-occupancy source need not admit a subpower decomposition of this kind. A finite model is the full grid (c_i, j_j) , $i, j \in \mathbf{Z}/R$, with deterministic joint selector

$$F_{\text{flag}}(c_i, j_j) = (i + j, i + j). \quad (105)$$

Every endpointwise factorizing mode is a matching, whereas the R level sets $i + j = s$ are such matchings. Hence its positive deterministic mode rank is R , by (202). Fractional positive splitting does not improve this number. With unit mode capacities, the R level sets give the primal upper bound; in the dual (205), assign weight $1/R$ to every one of the R^2 grid vertices. Every independent mode is a matching of size at most R , so this is dual feasible and has value R . Therefore

$$\boxed{\mathfrak{r}_+^{\text{mode}} = \kappa_+^{\text{mode}} = R.} \quad (106)$$

Both unlabelled marginals nevertheless have uniform loads. On an occupancy shell one may have $R \simeq a^{-1}$, reproducing the loss in (90). The motivic operation word and the structural q^2 provenance can remain fixed while this physical correlation rank grows.

Consequently the desired $2D/3D$ tensorization is proved on the external-product and fractional low-conflict flag sources. Its unresolved part is the same kind of correlated positive-selector/Hall obstruction as the one isolated in (206), now realized geometrically by intermediate anisotropic planks. This is a sufficient geometric realization of that interface, not an identification of the scalar and typed conditions. Marginal lower-dimensional Takeya estimates alone cannot bound it. The finite model does not assert failure of the geometric plank estimate; it proves only that no argument using the two marginal estimates plus positive (even fractional) coalescence can establish the general gate. A successful four-dimensional step must exploit additional joint plank geometry or assume the corresponding capacity.

Two genuine pieces of this interface do reduce to the three-dimensional output. In a factorized diffuse-fibre block, partition the cells into modes $\mathcal{C}_\eta$ so that, on the retained source,

$$0 \leq \tilde{\rho}_C \leq \rho_C, \quad \tilde{\rho}_C(z, t) = r_\eta(z, t) \quad (C \in \mathcal{C}_\eta), \quad \sum_\eta r_\eta(z, t) \leq K_f = \delta^{-o(1)}. \quad (107)$$

Writing $\widetilde{W}_{3,\eta} = \sum_{C \in \mathcal{C}_\eta} w_C$, one has

$$\widetilde{N}_4 = \sum_\eta r_\eta \widetilde{W}_{3,\eta} \leq K_f \widetilde{W}_3. \quad (108)$$

Hence the projected subsource (65) lifts immediately, with $\|\widetilde{N}_4\|_\infty \leq K_f K_p$. In the fibre-disjoint block, put $b_C = w_C \tilde{\rho}_C$, so that $\widetilde{N}_4 = \sum_C b_C$. If the one-cell pruning gives

$$\|b_C\|_\infty \leq K_0, \quad \sup_{z,t} \#\{C : b_C(z, t) > 0\} \leq Q, \quad K_0, Q = \delta^{-o(1)}, \quad (109)$$

then $\widetilde{N}_4 \leq K_0 Q$. These are respectively the scalar shadows of the external-product and conflict-free parts of the typed source below. They also satisfy (70), since a pointwise bound $\widetilde{N}_4 \leq B$ gives $\int \widetilde{N}_4^2 \leq B \int \widetilde{N}_4$.

They do not settle the correlated marked block: the latter is precisely the root collision (74), or more strongly the coherent atomwise capacity (67). Claiming that the ordinary three-dimensional sticky theorem already supplies either statement would hide the four-dimensional Takeya difficulty in the word “projection.”

8.8 Comparison with a stronger projection hypothesis

For comparison only, suppose one imposes the unavailable stronger condition that there are $B_C \subset A_C$ with

$$|B_C| \geq \lambda_{d-1} \sigma \delta^{d-2}, \quad \left\| \sum_C \mathbf{1}_{B_C} \right\|_\infty \leq B_{d-1}. \quad (110)$$

Then the following factorized fibre calculation is valid, but (54) shows that it is not a general route to (59).

Equations (47) and (110) imply

$$\int_{\pi^{-1}(B_C)} n_C = \int_{B_C} g_C \gtrsim \lambda \lambda_{d-1} m \delta^{d-1}. \quad (111)$$

Under the physical incidence law on this lifted set, the size-biased mean is bounded by

$$\frac{\int_{\pi^{-1}(B_C)} n_C^2}{\int_{\pi^{-1}(B_C)} n_C} \lesssim_d \frac{L}{\lambda \lambda_{d-1}}. \quad (112)$$

Hence, for every fixed $\zeta > 0$, Markov deletion of at most an $O_d(\delta^\zeta)$ fraction of the same physical shaded incidence mass gives

$$\boxed{n_C(x) \lesssim_{d,\zeta} \delta^{-\zeta} \frac{L}{\lambda \lambda_{d-1}}}. \quad (113)$$

There is no power of the cell occupancy m or of the projected speed σ .

For the subpower formulation, choose a slowly varying $\zeta = \zeta(\delta) \downarrow 0$ with $\zeta(\delta) \log(1/\delta) \rightarrow \infty$, slowly enough that the implicit constant in (113) is also $\delta^{-o(1)}$, and with δ^ζ below the per-level share of the deletion budget fixed below. Then the discarded fraction tends to zero and the cutoff loss is $\delta^{-o(1)}$. This is the only diagonal choice of exponents used below.

If the lower-dimensional retained multiplicity is

$$m_{d-1}(z) := \sum_C \mathbf{1}_{B_C}(z) \leq B_{d-1},$$

then partitioning the lifts by C gives the pointwise identity

$$m_d(x) \leq m_{d-1}(\pi x) \max_{C: \pi x \in B_C} n_C(x), \quad (114)$$

and consequently the rootwise cutoff

$$\boxed{B_d \lesssim_{d,\zeta} B_{d-1} \delta^{-\zeta} \frac{L}{\lambda \lambda_{d-1}}}. \quad (115)$$

All deletions here are measured in the ordinary physical incidence law and sum over the disjoint cells C .

Finally, if the rooted determinant moment satisfies $\mathbb{E}|s_1 \wedge \cdots \wedge s_{d-1}|^2 \geq a$, then

$$\Pr \{|s_1 \wedge \cdots \wedge s_{d-1}|^2 \geq a/2\} \geq \frac{a}{2-a} \geq \frac{a}{2}, \quad (116)$$

and every projected speed on that event is at least $(a/2)^{1/2}$. Thus when $a = \delta^{o(1)}$, determinant thresholding and the finitely many speed shells cost only source-subpower mass. Equation (46) and the planar estimate (53) close the projected marginal in dimension three. The lift to a three-dimensional multiplicity cutoff is exactly (59). The factorized shortcut (115) is valid only under the unavailable unweighted hypothesis (110), and is not used as an unconditional closure.

9 Filtered gluing on the global incidence source

9.1 The global first-failure decomposition

Fix a deletion budget $\tau(\delta) \rightarrow 0$ so slowly that $\tau(\delta)^{-1} = \delta^{-o(1)}$ and the sum of the budgets over all support, load and history levels still tends to zero. For any retained physical filtered stage X , let $m_X(x)$ be its Radon–Nikodym-corrected point multiplicity and define its hereditary cutoff by

$$B_X := \inf \left\{ B : \begin{array}{l} \text{there is a measurable hereditary set } E_X \subset \mathbf{R}^d \text{ with} \\ \int_{E_X^c} m_X \leq \tau(\delta) \int m_X, \quad m_X \mathbf{1}_{E_X} \leq B \text{ a.e.} \end{array} \right\}. \quad (117)$$

Thus every deletion is measured in the size-biased incidence mass of the same global source. Write λ_I for the hereditary shading-density parameter at support I ; at one dimensional handoff, $(\lambda_I, \lambda_{I \setminus \{0\}})$ is exactly the pair (λ, λ_{d-1}) in (112).

The preceding calculation is not exported as one certificate for each root. It realizes the fresh exact-support cofiber inside the same global filtered source. At support I , the physical realization of the Reedy triangle is

$$L_I^{\text{phys}} \longrightarrow K_I^{\text{phys}} \longrightarrow C_I^{\text{phys}}. \quad (118)$$

The first term is assembled from proper support faces, while the last term is the exact-support determinant block decomposed by (46)–(57). Its positive three-dimensional cutoff is the affine-fibre gate (59), not the projected unweighted shortcut. The chosen root describes only the local chart used to evaluate C_I^{phys} ; it does not replace (118) by a family of independently counted root outputs.

Concretely, use the single symmetric joint law (17). Every root contraction is a conditional kernel, but its target pushforward carries the density ρ_γ from (2). Normalization of the kernel does not permit one to erase this density. The one-cell matching example corresponds exactly to many histories with small chartwise multiplicity but a large summed ρ_γ -corrected target multiplicity.

The stable cofiber notation alone does not provide a positive measure decomposition, so one must choose it on the physical source. Fix the support order and the determinant thresholds used by the current branch. Apply Gram–Schmidt in that order to every packet: assign the packet to the first maximal subset $J \subseteq I$ whose declared pivots survive, using lexicographic tie breaking, and assign it to $J = I$ exactly when every pivot survives. Failed pivots are carried as the confluence/history fibre over J . This gives a disjoint measurable partition of the single joint packet law into its proper-face strata and its fresh exact-support stratum. Changing the support order only permutes finitely many such charts, and the symmetric law averages them without changing a one-point marginal.

Let $h_J(e)$ be the pushforward retention weight of an ordinary incidence atom e from the J -stratum. Because the strata partition one joint law,

$$0 \leq h_J(e) \leq 1, \quad \sum_{J \subseteq I} h_J(e) = 1 \quad (119)$$

before later branch deletion. Thus the support strata themselves do not duplicate an incidence. The history pushforward inside one stratum still has to be descended.

Dyadically filter a history block by its target density:

$$\Gamma_k := \{(\gamma, e) : 2^k \leq \rho_\gamma(e) < 2^{k+1}\},$$

and let $\text{Gr}_k^{\text{conf}} C_I^{\text{phys}}$ be the corresponding associated graded of (29). Finite geometric capacity gives only $O_d(\log(1/\delta))$ nonempty load shells. Put

$$\mathbf{N}_{I,k}(x) := \sum_{\gamma} \mathcal{E}_{\gamma}(\rho_{\gamma} \mathbf{1}_{\Gamma_k} f_{\gamma,I})(x), \quad (120)$$

where $f_{\gamma,I}$ is the shaded multiplicity observable on the exact-support I -block. The local calculations above estimate a fixed history chart. The required global confluence statement is the associated graded estimate

$$\boxed{\frac{\int \mathbf{N}_{I,k}(x)^2 dx}{\int \mathbf{N}_{I,k}(x) dx} \lesssim \delta^{-o(1)} \frac{L}{\lambda_I \lambda_{I \setminus \{0\}}} B_{K_{I \setminus \{0\}}},} \quad (121)$$

uniformly in k . This is R6. Its left side contains every cross-history term in the same load layer; proving only the fixed-root diagonal terms is not enough.

Put

$$A_I := \delta^{-o(1)} \frac{L}{\lambda_I \lambda_{I \setminus \{0\}}} B_{K_{I \setminus \{0\}}}.$$

On the event $\mathbf{N}_{I,k} > \tau(\delta)^{-1} A_I$, (121) gives

$$\int_{\{\mathbf{N}_{I,k} > \tau^{-1} A_I\}} \mathbf{N}_{I,k} \leq \frac{\tau}{A_I} \int \mathbf{N}_{I,k}^2 \leq \tau \int \mathbf{N}_{I,k}.$$

Thus, once (142) supplies R6, it produces exactly the hereditary cutoff of (117); the factor τ^{-1} is absorbed in $\delta^{-o(1)}$.

Once (121) holds, the confluence filtration and Cauchy–Schwarz over its $O(\log(1/\delta))$ graded layers give the support-step recurrence

$$\boxed{B_{K_I} \lesssim B_{L_I} + \delta^{-o(1)} \frac{L}{\lambda_I \lambda_{I \setminus \{0\}}} B_{K_{I \setminus \{0\}}}.} \quad (122)$$

This is a descent statement: the lower-support term, the confluence/load graded term and the successor are evaluated in one filtered object. It is not obtained by independently exporting a cutoff from each root.

9.2 Support recurrence and the source-mass ledger

The latching term has the same recombination issue. The first-failed-pivot rule of (42) assigns every packet to one proper face. If $m_{I \rightarrow J}$ is the multiplicity of the stratum assigned to $J \subsetneq I$, then

$$m_{L_I}(x) \leq \sum_{J \subsetneq I} m_{I \rightarrow J}(x).$$

There are only $O_d(1)$ support faces, but each face may contain many packet completions. Applying the lower-support cutoff independently to those packets would be the forbidden local-to-global step. They are therefore kept positively aggregated inside each structural history node, not made into additional vertices of the prefix tree below. The packet-carrier mark is used later only to disintegrate one common source before recombination. On the coherent narrow branch it coarsens to the unique cube label, and the entire coarse label alphabet has subpower size by (152).

The proper-predecessor charge controls all nonroot interactions, while (142) controls their root interaction. Conditional on that gate, the width, shell and final hardening losses are source-subpower, and the already established proper-face cutoffs give

$$\boxed{B_{L_I} \lesssim_d \delta^{-o(1)} \max_{J \subsetneq I} B_{K_J}.} \quad (123)$$

Substitution into (122) yields, under the same root-cross hypothesis, the closed support recurrence

$$\boxed{B_{K_I} \lesssim_d \delta^{-o(1)} \left(\max_{J \subsetneq I} B_{K_J} + \frac{L}{\lambda_I \lambda_{I \setminus \{0\}}} B_{K_{I \setminus \{0\}}} \right).} \quad (124)$$

Thus no unknown I -support quantity remains on the right once the physical root-cross energy is controlled. If the retained shading densities at the finitely many stages satisfy $\lambda_J \geq \delta^{o(1)}$, then $L = \delta^{-o(1)}$ and fixed-dimensional iteration gives

$$B_{K_I} \lesssim_d \delta^{-o(1)} B_{K_2}. \quad (125)$$

Without this density condition one keeps the displayed λ -factors; the descent does not erase sparse shading.

The required branch and density bounds follow from one source-mass ledger.

Across a fixed-dimensional descent, the first-failed-pivot partitions have $O_d(1)$ cells, while the speed, occupancy, target-density and confluence-load partitions each have $O_d(L)$ cells. Include the usual dyadic per-tube conditional-density regularization, but keep it distinct from selection of a tuple stratum. The cube-Gram split adds only its broad/narrow alternative. When a quantitative broad packet is retained, its conditional mass is at least $c_d \beta_{\text{Gr}}$ by (40), and its reciprocal is source-subpower. Quantizing a proper affine carrier at angular scale κ and offset scale $a_{\text{face}} \geq \kappa$ contributes at most $C_d \kappa^{-C_d}$ labels over the bounded physical region. Support drops at most $d - 2$ times, and the history tree is positively recombined rather than branch-selected. Hence only $O_d(1)$ dyadic partition types are selected along a branch, and the product of their cardinalities satisfies

$$\mathcal{P}_\delta \leq C_d L^{C_d} \beta_{\text{Gr}}^{-C_d} \kappa^{-C_d} = \delta^{-o(1)}. \quad (126)$$

Here is the normalization needed to compare the weighted Markov budgets with ordinary source mass. Every conditional history kernel has mass one, and integration of its Radon–Nikodym-corrected pushforward recovers the mass of its marked source. A fixed source atom occurs in only $O_d(1)$ reduced insertion orders; load shells partition rather than duplicate a history, and all selected chart refinements contribute at most a dimensional power of $\mathcal{P}_\delta$. Hence, if $\mathcal{I}_{\text{cur}}$ is the ordinary marked-incidence mass before the Markov cutoffs, then

$$\sum_{\text{retained stages } (I,k)} \int \mathbf{N}_{I,k}(x) dx \lesssim_d \mathcal{P}_\delta^{C_d} \mathcal{I}_{\text{cur}}. \quad (127)$$

By positivity, the ordinary marked-incidence mass removed at a stage is no larger than the corresponding removed filtered mass. Choose the diagonal budget above also so that $\tau(\delta) \mathcal{P}_\delta^{C_d} \rightarrow 0$. Summed over all stages, (127) then makes the ordinary source loss $o(\mathcal{I}_{\text{cur}})$.

For each tube choose a heaviest conditional-density cell and let $a(Q, T, x) \in \{0, 1\}$ be the resulting hereditary incidence mask. Put

$$\bar{a}_{Q,T} := \frac{1}{w_{Q,T}} \int_{Y_Q(T)} a(Q, T, x) dx$$

on the nonzero tube cells. Its ordinary one-slot retained mass is

$$\beta := \frac{1}{W} \sum_{Q,T} w_{Q,T} \bar{a}_{Q,T} \geq \mathcal{P}_\delta^{-1}.$$

Writing $r_Q = W_Q^{-1} \sum_T w_{Q,T} \bar{a}_{Q,T}$, the symmetric $|I|$ -slot packet mask has mass

$$\alpha_{\text{mask}} = \sum_Q \frac{W_Q}{W} r_Q^{|I|} \geq \left(\sum_Q \frac{W_Q}{W} r_Q \right)^{|I|} = \beta^{|I|}$$

by convexity. Tuple-stratum selection and the at most $d - 2$ support drops add only a dimensional power of $\mathcal{P}_\delta$. Consequently

$$\alpha \gtrsim_d \mathcal{P}_\delta^{-C_d} = \delta^{o(1)}, \quad \lambda_J^{-1} \lesssim_d \lambda_{\text{in}}^{-1} \mathcal{P}_\delta^{C_d} = \delta^{-o(1)} \quad (128)$$

whenever $\lambda_{\text{in}}^{-1} = \delta^{-o(1)}$. The global Markov deletions consume only an $o(1)$ fraction of ordinary source mass by (127), so they do not alter these bounds. Thus α and the later λ_J are outputs of the same source filtration, not independent assumptions.

9.3 Terminal unweighting

It remains to pass from the weighted descended source to an ordinary descendant once, at the end.

On the lifted $|I|$ -slot incidence law, let $A_{\text{tot}} \in [0, 1]$ be the product of all actual deletion and retention masks along the selected descent stratum; these masks are lifted to the marked incidences and symmetrized in the surviving slots. Confluence/history multiplicities are not included in A_{tot} ; they remain in the filtered observable m_{K_I} . Choose in (117) an admissible final cutoff event $G_I \subset \mathbf{R}^d$ with

$$m_{K_I} \mathbf{1}_{G_I} \leq 2B_{K_I},$$

and include $\prod_{j \in I} \mathbf{1}_{G_I}(x_j)$ in A_{tot} . The corresponding deletion has already been charged to the global budget.

Let Π_j be the j -th marked-incidence slot projection. Define the averaged slot density

$$h(e) := \frac{1}{|I|} \sum_{j \in I} \frac{d(\Pi_j)_*(A_{\text{tot}} \tilde{\mathbf{P}}_w)}{d\mu_{\text{inc}}}(e). \quad (129)$$

Every unmasked slot marginal of $\tilde{\mathbf{P}}_w$ is μ_{inc} , and $A_{\text{tot}} \leq 1$. Hence $0 \leq h \leq 1$, while

$$\int h d\mu_{\text{inc}} = \int A_{\text{tot}} d\tilde{\mathbf{P}}_w = \alpha.$$

Consequently

$$\int_{\{h < \alpha/2\}} h d\mu_{\text{inc}} \leq \alpha/2,$$

so, for $E = \{h \geq \alpha/2\}$,

$$\int_E h d\mu_{\text{inc}} \geq \frac{\alpha}{2}, \quad \mu_{\text{inc}}(E) \geq \int_E h d\mu_{\text{inc}}, \quad E \subseteq \hat{G}_I := \{(Q, T, x) : x \in G_I\}.$$

With m_h and m_E defined by (21), one has

$$\mu_{\text{inc}}(E) \geq \frac{\alpha}{2}, \quad m_E(x) \leq \frac{2}{\alpha} m_h(x) \leq \frac{2}{\alpha} m_{K_I}(x) \mathbf{1}_{G_I}(x) \leq \frac{4}{\alpha} B_{K_I}. \quad (130)$$

The last inequality follows from positivity and normalization of every history kernel: the filtered observable contains the retained incidence contribution and possibly additional history load. Therefore, after R6 has supplied the filtered weighted estimate, a descent with $\alpha = \delta^{o(1)}$ produces a genuine ordinary-incidence descendant with only $\delta^{-o(1)}$ loss. This single terminal threshold is the correct globalization; no intermediate root-certificate assembly is used.

10 Positive confluence recombination and the root gate

10.1 The history tree and least-common-ancestor charge

The confluence estimate is global, but its globalization is carried by the history filtration. The unified tower's reduced history normal form uses the terminal structural object once and retains every genuine intermediate frontier label once. Thus a marked reduced insertion chain is represented by a word

$$\gamma = (h_1, \dots, h_s; I),$$

in its actual insertion order, where I is written only once and no genuine intermediate frontier is duplicated. Different admissible orders remain different marked histories. Every prefix is again a legal reduced insertion history, and deleting its last label is its strict predecessor. This prefix

property, rather than an artificial global sorting of the labels, is the only part of the reduced normal form used below.

Arrange the retained words by common prefix and append a terminal marker to every complete word so that no leaf is a proper prefix of another. The retained histories on one load shell then form a finite rooted prefix tree $\mathcal{T}_{I,k}$: its leaves are the full histories, and every internal vertex is a strict confluence predecessor. A proper support face is removed before entering this tree and belongs to L_I^{phys} . The tree contains only these structural history labels. In particular the packet-carrier $\mathfrak{c}_{\text{pkt}}$ of (37) is not made into another tree level: all of its values remain positively aggregated inside the corresponding structural leaf. They are disintegrated only when the root-cross measure is analysed. This placement is essential; applying a predecessor cutoff separately to tiny carrier fragments would be the forbidden local-to-global argument.

On one confluence/load shell write

$$u_\gamma(x) := \mathcal{E}_\gamma(\rho_\gamma \mathbf{1}_{\Gamma_k} f_{\gamma,I})(x) \geq 0.$$

Then

$$\int \mathbb{N}_{I,k}^2 = \sum_{\gamma, \gamma'} \int u_\gamma(x) u_{\gamma'}(x) dx. \quad (131)$$

For a vertex a , let $\text{Desc}(a)$ be its descendant leaves and put

$$W_a(x) := \sum_{\gamma \in \text{Desc}(a)} u_\gamma(x). \quad (132)$$

The children of a partition its descendant leaves, hence

$$W_a = \sum_{b \in \text{ch}(a)} W_b. \quad (133)$$

This is an identity of nonnegative functions on the original incidence source, not a stable-category decomposition. The unified tower supplies only the legal prefix predecessors. Positivity and additivity here come from the single physical joint law: the cylinder sets of the child prefixes are disjoint and exhaust the parent cylinder, and their conditional kernels obey (14). More precisely, if a has confluence type $\chi(a) \prec \chi$, pushforward of that disjoint cylinder partition gives, for the descended shading f_a ,

$$W_a = \mathcal{E}_{\chi(a)}(f_a). \quad (134)$$

Thus an internal node is an observable in a strictly lower filtration stage, not a relabeling of the current full-history sum.

Let r_* be the root of $\mathcal{T}_{I,k}$. The valid induction invariant applies to every *proper* vertex $a \neq r_*$:

$$W_a(x) \leq B_I^{\text{pred}}, \quad B_I^{\text{pred}} := \delta^{-o(1)} \frac{L}{\lambda_I \lambda_{I \setminus \{0\}}} B_{K_{I \setminus \{0\}}},$$

after the usual source-small Markov deletion. For a leaf this is (113), with R5 absorbing its source-subpower Radon–Nikodym density. For a proper internal vertex it is the induction hypothesis applied to (134); a fresh Cartan vertex first passes to its root–Moore antecedent and then has smaller support. This uniform form is applied only to the whole retained structural node, whose total pushforward has source-subpower RN density. It is never invoked separately on its possibly power-many packet completions; those completions have already been summed in W_a , and their remaining pair interaction is treated at the root below.

The well-founded analytic order for these proper predecessors is

$$(|I|, \ell(\chi), n), \quad (135)$$

ordered lexicographically with support outermost, confluence length next and Cartan height last. A strict confluence predecessor lowers $\ell(\chi)$, a proper face lowers support, and a Cartan predecessor lowers n . Supports of size at most one are trivial latching faces. The minimal nontrivial leaf state is $(2, 0, 0)$, where the planar Córdoba estimate applies. This establishes the cutoff on proper vertices; it does not establish one on r_* , because

$$W_{r_*} = \sum_{\gamma} u_{\gamma} = \mathbf{N}_{I,k}. \quad (136)$$

Using the desired cutoff on r_* would therefore be circular.

For two leaves whose least common ancestor $a(\gamma, \gamma') \neq r_*$, define, with value zero when the denominator vanishes,

$$c_{\gamma, \gamma'}(x) := \frac{u_{\gamma}(x)u_{\gamma'}(x)}{W_{a(\gamma, \gamma')}(x)}. \quad (137)$$

The proper-node cutoff gives

$$u_{\gamma}u_{\gamma'} \leq B_I^{\text{pred}} c_{\gamma, \gamma'}. \quad (138)$$

For a proper internal vertex,

$$\sum_{a(\gamma, \gamma')=a} u_{\gamma}u_{\gamma'} = W_a^2 - \sum_{b \in \text{ch}(a)} W_b^2 \leq W_a^2,$$

and a leaf contributes its diagonal W_a^2 . Since the proper nodes at each depth have disjoint descendant sets,

$$\sum_{a(\gamma, \gamma') \neq r_*} c_{\gamma, \gamma'}(x) \leq H_{I,k} \mathbf{N}_{I,k}(x), \quad (139)$$

where $H_{I,k} = \delta^{-o(1)}$ is the refined tree height. Thus all proper-predecessor pairs satisfy

$$\sum_{a(\gamma, \gamma') \neq r_*} \int u_{\gamma}u_{\gamma'} \lesssim \delta^{-o(1)} B_I^{\text{pred}} \int \mathbf{N}_{I,k}.$$

What remains is exactly the cross term born at the root:

$$\mathbf{X}_{I,k}^{\text{root}} := \sum_{a(\gamma, \gamma')=r_*} \int u_{\gamma}u_{\gamma'} = \int \left(W_{r_*}^2 - \sum_{b \in \text{ch}(r_*)} W_b^2 \right). \quad (140)$$

Consequently the unconditional output of the tree calculation is

$$\int \mathbf{N}_{I,k}^2 \lesssim \delta^{-o(1)} B_I^{\text{pred}} \int \mathbf{N}_{I,k} + \mathbf{X}_{I,k}^{\text{root}}. \quad (141)$$

R6 follows precisely from the additional physical estimate

$$\boxed{\mathbf{X}_{I,k}^{\text{root}} \lesssim \delta^{-o(1)} B_I^{\text{pred}} \int \mathbf{N}_{I,k}.} \quad (142)$$

10.2 Fixed-carrier decomposition of the root term

The physical content of this gate can be separated without changing the history filtration. Refine each terminal history leaf by the ordinary tube to which its target incidence belongs. Positivity and the disjoint partition of the ordinary incidence source give

$$u_{\gamma} = \sum_T u_{\gamma, T}, \quad u_{\gamma, T} \geq 0, \quad (143)$$

where zero summands are omitted. This adds only a terminal label; it is not a new motivic coordinate and it does not duplicate physical mass. Hence the root term has the disjoint decomposition

$$\begin{aligned} \mathsf{X}_{I,k}^{\text{root}} &= \mathsf{X}_{I,k}^{\text{geom}} + \mathsf{X}_{I,k}^{\text{dup}}, \\ \mathsf{X}_{I,k}^{\text{geom}} &:= \sum_{\substack{a(\gamma,\gamma')=r_* \\ T \neq T'}} \int u_{\gamma,T} u_{\gamma',T'}, \\ \mathsf{X}_{I,k}^{\text{dup}} &:= \sum_{\substack{a(\gamma,\gamma')=r_* \\ T=T'}} \int u_{\gamma,T} u_{\gamma',T}. \end{aligned} \tag{144}$$

For a root child b define its terminal-tube load and the total terminal load by

$$D_{b,T}(x) := \sum_{\gamma \in \text{Desc}(b)} u_{\gamma,T}(x), \quad D_T(x) := \sum_{b \in \text{ch}(r_*)} D_{b,T}(x). \tag{145}$$

Then $\sum_T D_T = \mathsf{N}_{I,k}$, and expanding only pairs of distinct root children gives the exact positive identities

$$\mathsf{X}_{I,k}^{\text{dup}} = \int \sum_T \left(D_T^2 - \sum_b D_{b,T}^2 \right), \tag{146}$$

$$\mathsf{X}_{I,k}^{\text{geom}} = \int \sum_{T \neq T'} \left(D_T D_{T'} - \sum_b D_{b,T} D_{b,T'} \right). \tag{147}$$

Every bracket is nonnegative: it is the sum over $b \neq b'$. In particular, the same-terminal-tube subgate follows from the source-compatible hardening

$$\boxed{D_T(x) \lesssim \delta^{-o(1)} B_I^{\text{pred}} \quad \text{for every retained } (x, T),} \tag{148}$$

because

$$\mathsf{X}_{I,k}^{\text{dup}} \leq \int \sum_T D_T^2 \leq \|D_T\|_{L_{x,T}^\infty} \int \mathsf{N}_{I,k}.$$

Thus the duplicate branch asks for a terminal-incidence history-load theorem, not a tube-angle estimate and not a coefficient-depth lift.

Indeed, set

$$F_T(x) := \#\{b \in \text{ch}(r_*) : D_{b,T}(x) > 0\}.$$

Every root child retained in the hereditary branch-density range is a proper vertex, so $D_{b,T} \leq W_b \leq B_I^{\text{pred}}$ by the induction invariant. Therefore

$$D_T(x) \leq F_T(x) B_I^{\text{pred}}. \tag{149}$$

Consequently a source-compatible restriction with $F_T(x) \leq \delta^{-o(1)}$ proves (148). This is a sufficient raw-fragment criterion. The common-source argument below replaces the raw history count by an aggregate over the globally finite coarse-carrier alphabet; the needed cube coherence on the narrow branch comes from the physical Gram split, not from the formal tower.

10.3 Common-source aggregation

The full descent does remove multiplicity coming only from different history charts on one physical carrier. Let $\vartheta(b)$ be the typed predecessor of a root child, let $j(b) \in \{0, \dots, r\}$ record its terminal slot after the measure-preserving root exchange. The structural child b does not have one carrier label. Instead disintegrate its cylinder by the measurable packet mark:

$$C_{b,c} := C_b \cap \{\mathbf{c}_{\text{pkt}} = c\}, \quad W_b = \sum_c W_{b,c}, \quad D_{b,T} = \sum_c D_{b,c,T}.$$

For fixed (θ, j, c) , put

$$\overline{W}_{\theta,j,c} := \sum_{\substack{\vartheta(b)=\theta \\ j(b)=j}} W_{b,c}, \quad \overline{D}_{\theta,j,c,T} := \sum_{\substack{\vartheta(b)=\theta \\ j(b)=j}} D_{b,c,T}.$$

The sets $C_{b,c}$ are disjoint restrictions of $\tilde{\mathbf{P}}_{w,A}$, and for fixed (θ, j, c) every terminal map is the same coordinate projection Π_j . Hence, for every ordinary incidence set B ,

$$\begin{aligned} & \sum_{\substack{\vartheta(b)=\theta \\ j(b)=j}} (\Pi_j)_* (\mathbf{1}_{C_{b,c}} \tilde{\mathbf{P}}_{w,A})(B) \\ &= \sum_{\substack{\vartheta(b)=\theta \\ j(b)=j}} \tilde{\mathbf{P}}_{w,A}(C_{b,c} \cap \Pi_j^{-1}(B)) \leq \tilde{\mathbf{P}}_{w,A}(\Pi_j^{-1}(B)) \leq \alpha^{-1} \mu_{\text{inc}}(B). \end{aligned} \quad (150)$$

Thus (25) proves the aggregate comparison

$$\boxed{\frac{d \left(\sum_{\substack{\vartheta(b)=\theta \\ j(b)=j}} (\Pi_j)_* (\mathbf{1}_{C_{b,c}} \tilde{\mathbf{P}}_{w,A}) \right)}{d\mu_{\text{inc}}} \leq \alpha^{-1} = \delta^{-o(1)}}. \quad (151)$$

Let $\overline{W}_{\theta,j} := \sum_c \overline{W}_{\theta,j,c}$. This is one disjoint aggregate of proper structural nodes, so the predecessor invariant applies to $\overline{W}_{\theta,j}$; positivity then gives $\overline{W}_{\theta,j,c} \leq \overline{W}_{\theta,j} \lesssim B_I^{\text{pred}}$ without applying induction separately to the carrier fragment. This is the exact gain: charts sharing a typed predecessor and packet carrier are aggregated, but the carrier is not thereby made a function of the terminal incidence.

No entropy hypothesis is needed at the chosen carrier resolution. Let $\mathcal{C}_\kappa$ contain c_{br} and all quantized proper affine carriers occurring in (37), over the finitely many pivot positions and dimensions. A κ -net in each relevant projective Grassmannian has $O_d(\kappa^{-C_d})$ elements. Since the physical region is bounded and $a_{\text{face}} = \delta + \kappa \geq \kappa$, the corresponding affine offsets have the same type of bound. Thus

$$\boxed{|\mathcal{C}_\kappa| \leq C_d \kappa^{-C_d} = \delta^{-o(1)}}. \quad (152)$$

In particular, for a fixed terminal endpoint there are at most this many carrier marks produced by packet completions; arbitrarily many packets with the same mark remain in one common-source aggregate. Combining this count with (151) and the fixed number of typed predecessor/slot pairs gives directly

$$D_T(x) \leq \sum_{\theta,j,c} \overline{W}_{\theta,j,c}(x) \lesssim |\mathcal{C}_\kappa| B_I^{\text{pred}} \lesssim \delta^{-o(1)} B_I^{\text{pred}}. \quad (153)$$

This closes the duplicate subgate on both broad and narrow histories, without summing independent chartwise estimates.

On the intrinsic cube-narrow branch, (41) makes the carrier label $\mathfrak{c}_Q$ a function of Q . For each tube, partition its ordinary incidence mass among the labels in $\mathcal{C}_\kappa$. Given any slowly vanishing source-subpower τ , delete the fragments having conditional mass less than $\tau/|\mathcal{C}_\kappa|$. Their total mass on each tube, and hence globally, is at most τ , while every retained fragment has relative mass

$$\frac{\tau}{|\mathcal{C}_\kappa|} = \delta^{o(1)}. \quad (154)$$

The hereditary lower-dimensional estimate therefore applies to each fixed carrier with only subpower density loss. The carrier pieces partition the unary incidence mass; at each cube the label is unique, so their quadratic outputs are disjoint up to the bounded overlap of the cube decomposition. Consequently they recombine with at most the subpower factor in (152). Thus moving coarse carriers create no additional gate. This conclusion uses the scale $\kappa = \delta^{o(1)}$; it would be false for labels resolved at a fixed power of δ .

The first summand is interaction between distinct physical tubes. The second is repeated history load over the same unmarked terminal tube and has no angular separation. Thus a concrete sufficient, and branchwise natural, form of the root gate is

$$\boxed{\mathsf{X}_{I,k}^{\text{geom}} \lesssim \delta^{-o(1)} B_I^{\text{pred}} \int \mathsf{N}_{I,k}, \quad \mathsf{X}_{I,k}^{\text{dup}} \lesssim \delta^{-o(1)} B_I^{\text{pred}} \int \mathsf{N}_{I,k}.} \quad (155)$$

On one fixed projected line cell, the unweighted geometric kernel underlying the first estimate is the Córdoba calculation (48); after the load-shell and RN regularizations, this is the fixed-carrier input. Equations (152)–(154) permit its carrierwise application on the coherent cube-narrow branch. The second estimate is not a tube-intersection estimate: (150) and (153) close it by common-source aggregation.

The common-source aggregation is load-bearing. If one keeps only individual R5 bounds and forgets the disjoint cylinder source, a one-point model with R leaf histories, all having the same terminal tube, and $u_\gamma = 1$, every proper leaf has cutoff 1, but

$$\mathsf{N} = R, \quad \mathsf{X}^{\text{root}} = R^2 - R, \quad \frac{\int \mathsf{N}^2}{\int \mathsf{N}} = R.$$

Here $D_T = R$, so the failure lies in the duplicate subgate. This model cannot satisfy (150): R unit-mass children are not disjoint restrictions of one unit-mass common source. It shows why individual chartwise R5 is insufficient, not an obstruction to the fixed-carrier aggregate argument. Thus the duplicate subgate is closed globally, and the coherent cube-narrow part of the geometric subgate is also closed. What is not supplied by the carrier count is the positive assembly of broad exact-support pairs of distinct physical tubes into strict predecessors. The unified tower names those destinations, but the aggregate descent alone does not dominate their physical pair measure.

A rotating-star construction is an obstruction only to a much finer carrier alphabet. Indeed, one root tube may meet locally narrow stars whose normals visit power-many caps at angular resolution $\eta = \delta^c$ with fixed $c > 0$. This shows that local Gram narrowness does not control fine-scale normal variation. It does not obstruct the present decomposition: at resolution $\kappa = \delta^{o(1)}$, the whole normal-cap space, not merely the labels visited by one tube, has cardinality $\kappa^{-O_d(1)} = \delta^{-o(1)}$. Refining the tube-normal covariance into spectral ranks or principal-angle strata would therefore add non-load-bearing levels and is not used. The remaining issue is instead the broad exact-support pair assembly described next.

10.4 Positive routing to strict predecessors

For the full-motive route, a sufficient remaining interface needs no new descent coordinate. Remove the coherent cube-narrow part of the ordered terminal root-collision source, already

controlled by (153) and the fixed-carrier recombination following (154), and denote the broad exact-support event by $\mathcal{B}_{I,k}$. The cube-Gram alternative and the retained broad-packet stratum were fixed before the history tree was formed, so this is a measurable mark of each retained history; no unrecorded packet completion is being chosen here.

On a collision atom, $\mathbf{1}_{\mathcal{B}_{I,k}}$ means that the common cube $Q(x)$ is on the Gram-broad alternative and both histories carry the retained broad-packet mark. What is still required is a positive pair-realization into already existing strictly lower typed strata $\xi \prec (\eta; I, \chi, \nu, n)$. Put

$$d\Lambda_{I,k}^{\text{root}}(\gamma, T; \gamma', T'; x) := \mathbf{1}_{\mathcal{B}_{I,k}} \mathbf{1}_{\{T \neq T'\}} \mathbf{1}_{\{a(\gamma, \gamma') = r_*\}} u_{\gamma, T}(x) u_{\gamma', T'}(x) dx. \quad (156)$$

Write $\mathbf{X}_{I,k}^{\text{br}} := \|\Lambda_{I,k}^{\text{root}}\|$. A positive routing consists of kernels

$$\mathcal{R}_\xi(dh, dh' | z) \geq 0, \quad z = (\gamma, T; \gamma', T'; x), \quad \sum_\xi \mathcal{R}_\xi(1 | z) = 1$$

for $\Lambda_{I,k}^{\text{root}}$ -almost every z . Each kernel lands in a strict predecessor ξ , is the pair version of the R5-compatible unary maps, and preserves their common lifted physical point. It must vanish unless ξ occurs in the finite typed support of the algebraic routing (212). Put

$$\Lambda_\xi^{\text{root}}(A) := \int \mathcal{R}_\xi(A | z) d\Lambda_{I,k}^{\text{root}}(z).$$

Write $v_{\xi,h}(x) \geq 0$ for the complete RN-corrected terminal-history loads on the lifted predecessor and set

$$\mathbf{N}_\xi(x) := \sum_h v_{\xi,h}(x), \quad d\Lambda_\xi^{\text{pred}}(h, h'; x) := v_{\xi,h}(x) v_{\xi,h'}(x) dx. \quad (157)$$

Thus $\|\Lambda_\xi^{\text{pred}}\| = \int \mathbf{N}_\xi^2$. The exact remaining target is

$$\boxed{\Lambda_\xi^{\text{root}} \leq \delta^{-o(1)} \Lambda_\xi^{\text{pred}} \quad \text{for every strict } \xi.} \quad (158)$$

The kernels must be defined on the physical pair source and respect the support, confluence and Cartan predecessors supplied by the full motivic filtration; a stable-category decomposition alone is insufficient.

The unary destination ledger is already contained in the source-mass ledger, provided equal typed predecessors are coalesced rather than copied once for each source pair. Indeed, the complete loads $\mathbf{N}_\xi$ are existing retained predecessor stages, so (127) and (128) give

$$\sum_\xi \int \mathbf{N}_\xi \lesssim_d \mathcal{P}_\delta^{C_d} \mathcal{I}_{\text{cur}} \lesssim_d \mathcal{P}_\delta^{C_d} \alpha^{-1} \int \mathbf{N}_{I,k} = \delta^{-o(1)} \int \mathbf{N}_{I,k}. \quad (159)$$

Creating pair-dependent copies of the same ξ would invalidate this estimate and is not allowed. Since every ξ is strict, the already established predecessor R6 estimate gives

$$\|\Lambda_\xi^{\text{pred}}\| \lesssim B_I^{\text{pred}} \int \mathbf{N}_\xi.$$

Summing (158) gives

$$\mathbf{X}_{I,k}^{\text{br}} \lesssim \delta^{-o(1)} B_I^{\text{pred}} \int \mathbf{N}_{I,k},$$

which, together with the closed cube-narrow contribution, is the root gate.

10.5 The Hall capacity formulation

For fixed predecessor loads, kernel existence has an exact capacity formulation. Let $\mathcal{Z}_{I,k}^{\text{br}}$ be the source of (156), let $\mathcal{Y}^{\text{pred}} := \bigsqcup_{\xi} \mathcal{Y}_{\xi}^{[2]}$, and write $z \rightsquigarrow y$ when y is an allowed physical pair image of z in a typed predecessor belonging to (212). For $E \subseteq \mathcal{Z}_{I,k}^{\text{br}}$, put $\mathcal{A}_{\xi}(E) := \{y \in \mathcal{Y}_{\xi}^{[2]} : z \rightsquigarrow y \text{ for some } z \in E\}$. Then (158), with constant C_{δ} , is equivalent to the measurable Hall inequalities

$$\Lambda_{I,k}^{\text{root}}(E) \leq C_{\delta} \sum_{\xi} \Lambda_{\xi}^{\text{pred}}(\mathcal{A}_{\xi}(E)) \quad \text{for every measurable } E, \quad C_{\delta} = \delta^{-o(1)}. \quad (160)$$

Necessity follows by sending the routed mass of E into its allowed neighbourhood. For sufficiency, disintegrate over the preserved physical point x . At fixed δ , the tube, retained-history and destination alphabets in each fibre are finite. Equation (160) for all measurable E is therefore equivalent to its finite Hall cut inequalities for almost every x : if a fibre cut failed on a positive-measure set, one of the finitely many source label subsets would fail measurably there and would give a violating global E . On each fibre, max-flow/min-cut produces a flow with destination capacities $C_{\delta} \Lambda_{\xi,x}^{\text{pred}}$. Order the finite edges and choose the lexicographically first feasible basic flow; its entries are measurable functions of the fibre weights. Integrating these flows and disintegrating by the source weights gives the kernels $\mathcal{R}_{\xi}$.

Thus unary R5 bounds only special one-slot cuts; the unresolved analytic content is the full family of pair-source cuts in (160).

The single choice $E = \mathcal{Z}_{I,k}^{\text{br}}$, together with the unary ledger, is enough for the scalar root estimate. The reason to require every E is hereditary stability: subsequent support, shading and history restrictions replace the source by such subsets, and the same routing must remain valid without being reselected from target mass.

For a broad edge write

$$\mathcal{S}_{\text{alg}}(e, e') := \text{supp}_{\text{typ}} \mathfrak{R}_{\text{alg}}(e, e'),$$

the set of strict types having nonzero coefficient in the algebraic routing (212). The direct full-filtration route has a useful deterministic sufficient form which does not require a common unary predecessor. Let E_x^{br} be the finite once-oriented fibre of positive broad distinct-tube source pairs at x , with the fixed order on root children used to choose the orientation. Let

$$\Theta_x : E_x^{\text{br}} \longrightarrow \bigsqcup_{\xi} \mathcal{Y}_{\xi,x}^{[2]}, \quad \Theta_x(e, e') = (\xi, h, h'),$$

be a measurable joint lift of the first nonzero typed obstruction, with $\xi \in \mathcal{S}_{\text{alg}}(e, e')$. It may depend jointly on (e, e') .

Writing $u_e = u_{\gamma,T}$, it is enough to prove the weighted fibre bound

$$\sum_{\substack{(e,e') \in E_x^{\text{br}} \\ \Theta_x(e,e') = (\xi,h,h')}} u_e(x) u_{e'}(x) \leq C_{\delta} v_{\xi,h}(x) v_{\xi,h'}(x), \quad C_{\delta} = \delta^{-o(1)}, \quad (161)$$

for almost every x and every destination. Pushforward by Θ_x then gives (158) directly. This is a weighted physical codegree estimate for the joint FSC lift. Neither the raw cardinality of its fibre nor the structural q^2 confluence provenance proves (161). When no deterministic lift has this bound, the Hall formulation permits the source mass to split among several admissible destinations.

The distinction is already visible with a genuine product source. Give every $(i, j) \in (\mathbf{Z}/R)^2$ mass R^{-2} , and let a joint confluence map send

$$(i, j) \longmapsto (i + j, i + j).$$

Each target slot is still uniform, so both unary RN constants are one. The routed pair measure, however, has mass $1/R$ at each diagonal point, whereas the product of its unary target loads has mass $1/R^2$ there. The diagonal cut therefore forces $C_\delta \geq R$ in (160). Tensoring two unary R5 statements cannot control a predecessor map which depends jointly on the two endpoints; the missing input is precisely control of correlated confluence mass.

This example also marks the limit of the unary local-FSC data. If, after a strict type is selected, its physical pair map happens to be the external product of two existing unary R5 maps, positivity and Fubini give the required pair domination. But the root-cross source is defined by $a(\gamma, \gamma') = r_*$: the two histories have no common strict prefix other than the current root, whose cutoff is the assertion being proved. Sending both endpoints to an arbitrary proper support face is no substitute, because the broad exact-support determinant dies there. Thus neither unary R5 nor the structural pullback compatibility of the all-support comparison supplies the required pair lift. The general route must prove the joint weighted-fibre estimate (161), or, more generally, the splittable Hall estimate (160); no separate unary-factorization hypothesis is imposed.

10.6 The scalar cycle-space reduction

There is a sharper scalar cycle-space reduction. At a fixed x , let the vertices be the active children $b \in \text{ch}(r_*)$. Write $U_b(x)$ for the broad endpoint load in that child and, for $b \neq b'$, let $M_{bb'}(x)$ be the corresponding unordered broad distinct-tube collision mass; explicitly,

$$U_b(x) := \sum_{\gamma(e) \in \text{Desc}(b)} \mathbf{1}_{\text{br}}(e; x) u_e(x),$$

$$M_{bb'}(x) := \sum_{\substack{\gamma(e) \in \text{Desc}(b), \gamma(e') \in \text{Desc}(b') \\ T(e) \neq T(e')}} \mathbf{1}_{\text{br}}(e; x) \mathbf{1}_{\text{br}}(e'; x) u_e(x) u_{e'}(x).$$

where $\mathbf{1}_{\text{br}}$ is the broad mark already fixed in (156). Then

$$0 \leq M_{bb'}(x) \leq U_b(x) U_{b'}(x), \quad U_b(x) \leq W_b(x) \leq B_I^{\text{pred}}. \quad (162)$$

Let G_x^{br} be the graph with an edge $\{b, b'\}$ precisely when $M_{bb'}(x) > 0$. For a spanning forest F_x , orient every tree away from one root. Each nonroot vertex is then charged once, so

$$\sum_{\{b, b'\} \in F_x} M_{bb'}(x) \leq B_I^{\text{pred}} \sum_b U_b(x).$$

Thus the only uncharged mass lies on the independent chords. Define the weighted cycle excess

$$\mathfrak{C}_{I,k}^{\text{br}} := 2 \int_{F_x} \min_{\text{spanning forest}} \sum_{\{b, b'\} \in E(G_x^{\text{br}}) \setminus E(F_x)} M_{bb'}(x) dx. \quad (163)$$

The finite graph permits a measurable lexicographic choice of a minimizing forest, and the preceding charge gives

$$\boxed{X_{I,k}^{\text{br}} \leq 2B_I^{\text{pred}} \int \mathbf{N}_{I,k} + \mathfrak{C}_{I,k}^{\text{br}}.} \quad (164)$$

Consequently, the scalar root gate closes if, at every retained support and shading restriction, one has

$$\boxed{\mathfrak{C}_{I,k}^{\text{br}} \leq \delta^{-o(1)} B_I^{\text{pred}} \int \mathbf{N}_{I,k}} \quad (165)$$

A coarser sufficient condition is

$$\text{ess sup}_x \beta_1(G_x^{\text{br}}) \leq \delta^{-o(1)}, \quad \beta_1(G) = |E(G)| - |V(G)| + c(G), \quad (166)$$

because a spanning forest has exactly $\beta_1(G_x^{\text{br}})$ chords and each chord is bounded by $B_I^{\text{pred}} \sum_b U_b(x)$. The weighted form (165) is preferable: it allows many cycles provided their physical load is small. The raw cycle rank is monotone under inherited graph deletion. The normalized weighted budget must be verified uniformly after every retained restriction, just like the root gate itself.

The forest decomposition is a localization of the difficulty, not a cancellation theorem. For example, suppose at one x there are n active children, $U_b = a = B_I^{\text{pred}}$, and their terminal-tube supports are pairwise disjoint. Then $G_x^{\text{br}} = K_n$, $M_{bb'} = a^2$, and

$$2 \min_{F_x} \sum_{E(K_n) \setminus E(F_x)} M_{bb'} = (n-1)(n-2)a^2,$$

whereas the whole ordered root-cross mass is $n(n-1)a^2$. Thus a spanning forest removes only $O(n)a^2$ from an $O(n^2)a^2$ interaction. Any useful cycle budget must come from independent geometric sparsity or small chord weights; selecting an optimal forest does not create it.

Pairwise tube geometry alone cannot supply that sparsity in dimensions $d \geq 3$. Take a direction-separated bush of $N \simeq \delta^{-(d-1)}$ unit δ -tubes through one common δ -ball and write $m = \sum_T \mathbf{1}_T$. The angular-shell count

$$\#\{T' : \angle(T, T') \simeq \theta\} \lesssim (\theta/\delta)^{d-1}, \quad |T \cap T'| \lesssim \frac{\delta^d}{\theta},$$

gives a contribution $O(\delta\theta^{d-2})$ per dyadic shell and per fixed T . Its shell sum is $O(\delta)$ for $d \geq 3$, whereas it is $O(\delta \log(1/\delta))$ for $d = 2$. In $d \geq 3$, the common ball contributes $N^2\delta^d \simeq N\delta$ after summation and gives the matching lower order. Thus

$$\boxed{\int m \simeq N\delta^{d-1} \simeq 1, \quad \int m^2 \simeq N\delta \simeq \delta^{-(d-2)} \quad (d \geq 3).} \quad (167)$$

In $d = 2$ the same upper-bound calculation loses only a logarithm, which is the Córdoba regime [2]; in $d \geq 3$ the loss is a genuine power and comes from broad angular shells. Thus neither direction separation nor a repeated pairwise planar estimate proves (165) on the unpruned source. Deleting only the common δ -ball does not fix this: the outer angular shells still have large multiplicity. A single bush can nevertheless be pruned cheaply. If $R_\delta \rightarrow 0$ and $R_\delta^{-1} = \delta^{-o(1)}$, delete the concentric R_δ -ball. This loses only an $O(R_\delta)$ -fraction of every tube, while direction packing gives

$$m(x) \lesssim (|x - x_0| + \delta)^{-(d-1)} \lesssim R_\delta^{-(d-1)} = \delta^{-o(1)} \quad (|x - x_0| \geq R_\delta).$$

What is missing in general is therefore a single source-compatible pruning or redistribution of all dense cycle cores. Precisely, on the current marked incidence source one would need one measurable mask H , with

$$\boxed{\begin{aligned} \mu_{\text{inc}}(H) &\geq \delta^{o(1)} \mu_{\text{inc}}(\text{current source}), \\ \mathfrak{C}_{I,k}^{\text{br}}[a] &\leq \delta^{-o(1)} B_I^{\text{pred}}[a] \int \mathbf{N}_{I,k}[a] \quad \text{for every } 0 \leq a \leq \mathbf{1}_H. \end{aligned}} \quad (168)$$

Here brackets mean the inherited restriction of every source, history and predecessor load by the same mask. This is one condition on all later restrictions, not a new mask chosen for each bad core. If (168) were proved at each of the finitely many descent stages, the source ledger would retain subpower incidence mass and (165) would close the scalar proof. For the hereditary typed route, the corresponding exact target is the all-subset Hall condition (160) on the source induced by H . The bush shows why deleting one visible core can be cheap; pairwise geometry gives no simultaneous choice of H for all overlapping cores. Selecting a different deletion separately for each core would therefore not suffice.

10.7 Pruning, bushes, and the maximal-function alternative

The direct pruning attempts stop at exactly this point. Bernoulli thinning of root children does not help: in the complete-graph example above, retaining a fraction p leaves about pn active children and cycle ratio of order pn , so a subpower ratio forces $p \lesssim \delta^{-o(1)}/n$; for power-sized n this discards a power fraction of the incidence source. Deleting the region of the currently worst core is cheaper for a single bush, but is not hereditary: a later restriction may expose a different dense subcore. Iterating that deletion would require a packing estimate of the form

$$\mu_{\text{inc}}\left(\bigcup_{\mathcal{C}}\{(T, x) : x \in \Omega_{\mathcal{C}}, T \in V(\mathcal{C})\}\right) \leq \tau(\delta)\mu_{\text{inc}}(\text{current source}), \quad \tau(\delta) \rightarrow 0, \quad (169)$$

10.8 Average carrier packing

The correct quantity for a multi-bush system is an average tube-core Carleson charge, not a uniform bound for every tube. Index the selected cores by j , let Ω_j lie in a ball of radius $R_j \geq \delta$, and let V_j be the tubes charged to that core. If every current shading has density at least λ , define

$$\mathfrak{b}(T) := \min\left\{1, C\lambda^{-1} \sum_{j: T \in V_j} R_j\right\}. \quad (170)$$

A unit δ -tube meets an R_j -ball in volume $O(R_j\delta^2)$. Subadditivity of the union, followed by summation in the ordinary shaded incidence law, therefore gives

$$\frac{\mu_{\text{inc}}\left(\bigcup_j\{(T, x) : T \in V_j, x \in \Omega_j\}\right)}{\mu_{\text{inc}}(\text{current source})} \leq \frac{\sum_T |Y(T)| \mathfrak{b}(T)}{\sum_T |Y(T)|}. \quad (171)$$

Thus (169) follows from smallness of the average on the right; a few repeatedly charged spine tubes are allowed.

This closes the canonical multi-bush test on the current per-tube shading shell. Suppose M disjoint R -balls support q -bushes, all bushes share one spine tube, and all other spokes occur in only one bush. There are $1 + M(q - 1)$ tubes. The spine contributes at most one full tube of source mass, while the spokes lose at most $O(R/\lambda)$ each. Hence

$$\frac{\sum_T |Y(T)| \mathfrak{b}(T)}{\sum_T |Y(T)|} \lesssim \frac{R}{\lambda} + \frac{1}{Mq}. \quad (172)$$

Choose the cutoff K slowly enough that the broad-bush deletion radius $R = K^{-1/2}$ satisfies $R/\lambda \rightarrow 0$; this is compatible with $K = \delta^{-o(1)}$. Then arbitrarily many bushes on one spine are harmless in the global source ledger. The earlier objection based on the worst tube was therefore too strong.

For a general system, first split the radii dyadically and write

$$I_R := \sum_{j: R_j \simeq R} |V_j| = \sum_T d_R(T), \quad d_R(T) := |\{j : R_j \simeq R, T \in V_j\}|. \quad (173)$$

On the same per-tube shading shell, $|Y(T)|$ is comparable across active tubes. Using $\min(1, u) \leq u$ in (170), the contribution of the R -shell to the average in (171) is at most

$$\boxed{\bar{\mathfrak{b}}_R \lesssim \min\left\{1, \frac{RI_R}{\lambda N}\right\}}, \quad N := |\mathbb{T}|. \quad (174)$$

Consequently every collection of non-saturated shells satisfying

$$\sum_R \frac{RI_R}{\lambda N} = o(1) \quad (175)$$

already obeys (169). This conclusion is purely geometric and uses no sticky theorem. Failure can occur only on scales for which

$$I_R \gtrsim \frac{\lambda N}{R}. \quad (176)$$

Since a unit tube meets at most $O(R^{-1})$ disjoint R -balls, the last display says that a positive source fraction is close to saturating its available longitudinal core slots. This is the precise carrier-tree meaning of stickiness in the present source ledger.

The same argument does not close a general reused-spoke network. Abstractly, take a bipartite incidence graph between tubes and cores in which every core has q tubes and every tube belongs to $D \simeq \lambda/R$ cores. Then $\mathfrak{b}(T) \simeq 1$ for essentially every tube, independently of how large the graph cycle rank is. Consequently neither cycle rank nor the number of cores bounds the right side of (171). Geometry must rule out, factor, or estimate this regular reuse pattern.

In tube language that pattern is the sticky alternative: the same source factors through nested thick carriers at many scales. A multiscale Frostman/Katz–Tao nonconcentration estimate [5] bounds the average charge in the non-sticky branches; when the estimate persists at every scale, the required terminal multiplicity bound is exactly the sticky Makeya input [7, 10], which in the present normalization is

$$\boxed{|U(\mathbb{T}, Y)| \geq \delta^{o(1)} \sum_{T \in \mathbb{T}} |Y(T)|, \quad \mu(\mathbb{T}, Y) \leq \delta^{-o(1)}}. \quad (177)$$

We invoke at the same time the sticky/non-sticky factoring theorem from [10]. Its multiplicity factorization is performed before passing to terminal carrier families and therefore coalesces overlapping physical images; it is not a sum of independently chosen local masks. The ball-like non-saturated factors are charged by (174); the reduction’s slab and plank factors are controlled by its stated multiplicity estimates; and every all-scale sticky terminal factor is covered by (177).

Indeed, put $m = \sum_T \mathbf{1}_{Y(T)}$, $\bar{m} = \int m/|U|$, and $H_{\text{st}} := \{x \in U : m(x) \leq 2\bar{m}\}$. Markov’s inequality in Lebesgue measure and the fact that $m \geq 1$ on U give

$$|H_{\text{st}}| \geq \frac{1}{2}|U|, \quad \int_{H_{\text{st}}} m \geq |H_{\text{st}}| \geq \frac{1}{2}\delta^{o(1)} \int m, \quad \|m\mathbf{1}_{H_{\text{st}}}\|_{\infty} \leq 2\delta^{-o(1)}.$$

Thus the low-multiplicity mask gives (59). This closes the scalar three-dimensional affine-fibre gate. The imported terminal estimate is not a consequence of (52) and is not used for the typed Hall routing.

The motivic carrier history records the factorization and makes the average in (171) hereditary, but it supplies no positive estimate for the regular sticky network. Thus the multi-bush calculation removes spine reuse as an obstruction and isolates the remaining three-dimensional input to sticky carrier packing, rather than to cycle rank alone.

Here the family of regions must be source-compatible and must hit every inherited $\delta^{-o(1)}$ -dense cycle core $\mathcal{C}$. The two-tube intersection bound controls one selected $\Omega_{\mathcal{C}}$, as in the bush, but does not sum (169) over cores met successively by the same tube. A disjoint-ball selection does not repair the sum: at radius R , one unit tube may meet $O(R^{-1})$ disjoint selected balls and hence lose $O(1)$, rather than $o(1)$, of its length. Thus the greedy proof of (168) needs a new Carleson-type packing input; neither the source ledger nor planar Córdoba supplies it.

10.9 Maximal and three-dimensional set inputs

There is, however, a clean conditional closure by the Kakeya maximal conjecture; see [4, 5, 11] for the classical maximal-function framework and its relation to the set problem. Put $p_d = d/(d-1)$. After the existing load and per-tube shading regularizations, its weighted hereditary form says that, for every current direction-separated terminal shell and every collection $0 \leq f_T \leq \mathbf{1}_{Y(T)}$,

$$\boxed{\int \left(\sum_T f_T \right)^{p_d} \leq K_\delta \int \sum_T f_T, \quad K_\delta = \delta^{-o(1)}. \quad (178)}$$

This is the usual conjectural maximal estimate in the form needed here. To pass from tube indicators to (178), decompose the coefficients dyadically. On a retained cell $f_T \simeq a \mathbf{1}_{Y(T)}$, the ordinary maximal bound is applied to the active tubes, while $|Y(T)| \geq \lambda |T|$, $\lambda^{-1} = \delta^{-o(1)}$, converts their total tube volume to at most $\lambda^{-1} \sum_T |Y(T)|$; the factors $a^{p_d} \leq a$ and the subpower number of load cells give the displayed form.

Indeed, (153) gives, on the current shell,

$$0 \leq D_T(x) \leq A_\delta B_I^{\text{pred}} \mathbf{1}_{Y(T)}(x), \quad A_\delta = \delta^{-o(1)}.$$

Set

$$f_T := \frac{D_T}{A_\delta B_I^{\text{pred}}}, \quad M := \sum_T f_T,$$

and fix the deletion budget $\tau(\delta)$ from the source ledger. With

$$L_\delta := \left(\frac{K_\delta}{\tau(\delta)} \right)^{1/(p_d-1)} = \left(\frac{K_\delta}{\tau(\delta)} \right)^{d-1} = \delta^{-o(1)}, \quad H := \{M \leq L_\delta\}, \quad (179)$$

the maximal estimate and size-biased Markov inequality give

$$\int_{H^c} M \leq L_\delta^{1-p_d} \int M^{p_d} \leq \tau(\delta) \int M. \quad (180)$$

Thus this single point mask retains $1 - \tau(\delta)$ of the current terminal load; (127) places its loss inside the same ordinary physical source-deletion budget. On H ,

$$\mathbf{N}_{I,k} = \sum_T D_T \leq A_\delta L_\delta B_I^{\text{pred}} = \delta^{-o(1)} B_I^{\text{pred}}.$$

Every later mask $0 \leq a \leq \mathbf{1}_H$ only decreases this positive load, so

$$\boxed{\mathbf{X}_{I,k}^{\text{root}}[a] \leq \int \mathbf{N}_{I,k}[a]^2 \leq \delta^{-o(1)} B_I^{\text{pred}} \int \mathbf{N}_{I,k}[a]. \quad (181)}$$

Consequently the Kakeya maximal conjecture implies the hereditary scalar root gate, (165), and (168); no core-by-core deletion is needed. This implication is scalar. It gives neither a common-point typed lift nor the destination-by-destination cuts (160).

The distinction from the recent three-dimensional result is essential. Wang–Zahl prove the Kakeya *set* conjecture in $\mathbb{R}^3$ [6]; this does not establish the Kakeya maximal function conjecture, which remains a separate stronger statement [11]. Hence their theorem cannot be substituted for (178) in the near-full-mass argument above.

It does nevertheless suffice for the weaker retention required by the present source ledger in dimension three. In the shaded form of the Wang–Zahl theorem, stated explicitly in [10], every active direction-separated family with shading density $\delta^{o(1)}$ satisfies

$$|U(\mathbb{T}, Y)| \geq S_\delta |\mathbb{T}| |T|, \quad S_\delta = \delta^{o(1)}. \quad (182)$$

Here $U(\mathbb{T}, Y) := \bigcup_{T \in \mathbb{T}} Y(T)$. Work on one of the existing dyadic load cells, so that $f_T \simeq a \mathbf{1}_{Y(T)}$, and put

$$M := \sum_T f_T, \quad I := \int M, \quad U := U(\mathbb{T}, Y), \quad \overline{M} := I/|U|.$$

Since $I \leq a|\mathbb{T}||T|$, (182) gives $\overline{M} \leq aS_\delta^{-1} = \delta^{-o(1)}$. Lebesgue Markov inequality and the lower bound $M \gtrsim a$ on U show that the single mask

$$H_{WZ} := \{x \in U : M(x) \leq 2\overline{M}\} \quad (183)$$

satisfies

$$|H_{WZ}| \geq \tfrac{1}{2}|U|, \quad \int_{H_{WZ}} M \gtrsim S_\delta I, \quad M \mathbf{1}_{H_{WZ}} \leq 2aS_\delta^{-1} = \delta^{-o(1)}. \quad (184)$$

Thus the mask retains the source-subpower fraction required by the ledger, and the normalization by $A_\delta B_I^{\text{pred}}$ used above gives (181) for this retained source and all its later restrictions. Consequently the Wang–Zahl set theorem closes the scalar physical gate in $d = 3$, despite not proving the maximal conjecture. This is an import of the full three-dimensionalakeya set theorem, not a derivation from planar Córdoba; it neither closes dimensions $d \geq 4$ nor supplies the typed Hall routing.

At scalar subpower resolution, the exact content can therefore be compressed to one refinement statement. On every regularized weighted tube cell there must be a single point mask H such that

$$\boxed{\int_H M \geq s_\delta \int M, \quad \|M \mathbf{1}_H\|_\infty \leq K_\delta, \quad s_\delta^{-1}, K_\delta = \delta^{-o(1)}}. \quad (185)$$

For a dyadic cell this is equivalent, up to subpower factors, to the shadedakeya union-volume statement. Indeed, (185) implies

$$|U| \geq |\text{supp}(M \mathbf{1}_H)| \geq \frac{\int_H M}{\|M \mathbf{1}_H\|_\infty} \geq \delta^{o(1)} \int M.$$

Conversely, a union lower bound and the average-multiplicity selection (183)–(184) give (185). Normalizing D_T by $A_\delta B_I^{\text{pred}}$ then gives (181). Thus the scalar pruning needed in this paper has the strength of theakeya set conjecture, not the stronger maximal conjecture; the latter improves the retained fraction from subpower to $1 - o(1)$.

This also prevents an automatic dimension lift. The proper-face/narrow branch does descend from d to $d - 1$, but the broad exact-support source remains in dimension d . Applying the three-dimensional theorem at the bottom therefore leaves, already in $d = 4$, precisely (185) for the four-dimensional broad source. Proving that step is the four-dimensionalakeya set problem in this language; it is not supplied by the three-dimensional base or by formal support descent. Even the narrow branch requires the stronger hereditary fibre-aware descendant (59); an unweighted projected multi-translation predicate is false by (54). Thus any upward recursion must carry the affine fibre coordinate and its confluence history rather than use a bare lower-dimensional union estimate.

Nor can the all-support faces be glued by projection alone. If π_j deletes the j -th coordinate, Loomis–Whitney has the direction

$$|E|^{d-1} \leq \prod_{j=1}^d |\pi_j E|;$$

lower bounds for all face projections give no lower bound for $|E|$. For example, inside a fixed box the slab

$$E_\delta := \{x : |x_1 + \cdots + x_d| \leq \delta\}$$

has $|E_\delta| \simeq \delta$, while every coordinate hyperplane projection has volume bounded below by a dimensional constant. This example only tests formal face gluing, not a broad tube theorem, but it identifies the missing datum: a reverse passage needs a uniform bound on the load in the common projection fibres. In the present construction that datum is exactly the physical fibre cutoff/root-cross capacity, not an all-support motivic identity. Exploiting broad transversality to obtain such a reverse estimate would require an additional multilinear Takeya or Brascamp–Lieb type inequality; it is not a consequence of the lower-dimensional set theorem.

10.10 Pointwise broadness and forest congestion

There is a second obstruction before such a multilinear estimate can even be applied. Equation (39) is an average over the incidence tuple law in a cube Q ; the root collision measure is supported on a common physical point x . Define the pointwise direction Gram operator, when the denominator is nonzero, by

$$G_{Q,x} := \frac{\sum_T \mathbf{1}_{Y_Q(T)}(x) e_T e_T^\top}{\sum_T \mathbf{1}_{Y_Q(T)}(x)}.$$

A broad averaged operator G_Q need not give positive mass to points where $G_{Q,x}$ is broad. Partition a cube into d disjoint measurable regions and shade only one of d transverse direction classes on each region. The integrated direction law can have $\det G_Q \simeq 1$, while $\det G_{Q,x} = 0$ almost everywhere. Thus the retained packet event (40) does not by itself imply the pointwise broad inequality required by multilinear Takeya.

The missing transfer would have to produce one retained point set E_Q with

$$\int_{E_Q} \sum_T \mathbf{1}_{Y_Q(T)} \geq \delta^{o(1)} \int_Q \sum_T \mathbf{1}_{Y_Q(T)}, \quad \det G_{Q,x} \geq \delta^{o(1)} \quad (x \in E_Q), \quad (186)$$

or else place the complementary pointwise-narrow mass in one coherent carrier field. The displayed disjoint-shading example shows that the first alternative alone is false. Choosing a different normal on each narrow point returns to the forbidden local-to-global assembly; proving a coherent selection with source-subpower loss is another form of the physical packing problem. Hence endpoint multilinear Takeya can estimate a genuinely pointwise-broad descendant, but the existing cube-average motivic mark does not construct that descendant.

A sharper scalar condition measures weighted reuse of a forest rather than the number of chords. Let F_x° be any spanning forest of G_x^{br} , put

$$\mathcal{C}_x(F^\circ) := E(G_x^{\text{br}}) \setminus E(F_x^\circ),$$

write $M_c := M_{bb'}$ for $c = \{b, b'\}$, and for $c \in \mathcal{C}_x(F^\circ)$ let $P_c \subseteq F_x^\circ$ be its unique fundamental path. Split the chord weight among its path edges by numbers

$$\alpha_{c,f}(x) \geq 0, \quad \alpha_{c,f} = 0 \text{ if } f \notin P_c, \quad \sum_{f \in P_c} \alpha_{c,f} = 1.$$

Define the weighted fundamental-cycle congestion, with value zero when there are no chords, by

$$\kappa_{F^\circ}(x) := \inf_{\alpha} \max_{f \in F_x^\circ} \frac{\sum_{\substack{c \in \mathcal{C}_x(F^\circ) \\ f \in P_c}} \alpha_{c,f}(x) M_c(x)}{M_f(x)}. \quad (187)$$

Every forest edge has positive weight by the definition of G_x^{br} . Finite weighted Hall duality gives the exact cut formula

$$\kappa_{F^\circ}(x) = \max_{\emptyset \neq A \subseteq \mathcal{C}_x(F^\circ)} \frac{\sum_{c \in A} M_c(x)}{\sum_{f \in \bigcup_{c \in A} P_c} M_f(x)}. \quad (188)$$

Indeed, the chord edges are demands, the forest edges have capacities $K M_f(x)$, and a chord is adjacent precisely to the edges of its fundamental path. The cut maximum has a connected-core form. If $\mathcal{H}(F_x^\circ)$ denotes the nonempty connected subtrees of F_x° , then

$$\kappa_{F^\circ}(x) = \max_{H \in \mathcal{H}(F_x^\circ)} \frac{\sum_{\substack{c \in \mathcal{C}_x(F^\circ) \\ P_c \subseteq H}} M_c(x)}{\sum_{f \in H} M_f(x)}. \quad (189)$$

To see this, start with a chord cut A , replace it by all chords whose paths lie in $\bigcup_{c \in A} P_c$, and split that path union into connected components. Its ratio is a weighted average of the component ratios, so one component is at least as large. The converse follows by taking all chords whose paths lie in a fixed H .

Consequently a failure at level K has a measurable connected witness H_x satisfying

$$\sum_{\substack{c \in \mathcal{C}_x(F^\circ) \\ P_c \subseteq H_x}} M_c(x) > K \sum_{f \in H_x} M_f(x). \quad (190)$$

Choose the lexicographically first witness among those with the fewest forest edges. Every edge of this minimal H_x lies on a retained chord path: otherwise delete that edge, split into components, and retain a component whose ratio is still larger than K , contradicting minimality. Hence every forest edge belongs to the fundamental cycle of some retained chord, and the graph formed by H_x and those chords has no bridge. We call this the weighted physical cycle core. This extraction is finite and measurable; it creates neither a new descent level nor an estimate.

Finally,

$$\sum_{c \in \mathcal{C}_x(F^\circ)} M_c = \sum_{f \in F_x^\circ} \sum_c \alpha_{c,f} M_c \leq \kappa_{F^\circ}(x) \sum_{f \in F_x^\circ} M_f.$$

Minimizing over the finite set of spanning forests gives a measurable quantity

$$\kappa_{\text{cyc}}(x) := \min_{F_x^\circ} \kappa_{F^\circ}(x).$$

Consequently the hereditary condition

$$\text{ess sup}_x \kappa_{\text{cyc}}(x) \leq \delta^{-o(1)} \quad (191)$$

implies (165): the minimum in (163) is no larger than the chord mass for a forest minimizing κ_{F° , while that forest's mass is at most $B_I^{\text{pred}} \sum_b U_b(x)$. Unlike (166), this condition permits many chords when their demands can be spread across sufficiently large path weights. It is a scalar graph criterion only; it does not construct a faithfully labelled motivic or physical predecessor lift.

For the full-motive route the forest must satisfy an additional compatibility which is invisible in the scalar graph. Call a spanning forest F_x *motivic-admissible* if, for every physical chord

atom z with child edge c , the existing operation-history pair complex contains a two-chain $\Sigma_{z,F}^{\text{mot}}$ whose structural boundary satisfies

$$\boxed{\pi_{\text{alg}} \partial_{\text{mot}} \Sigma_{z,F}^{\text{mot}} = [c] - \sum_{f \in P_F(c)} \epsilon_{c,f} [f],} \quad (192)$$

where π_{alg} forgets the operation-history labels down to the root-child graph, and with every remaining uncompressed boundary face lying in a strict typed predecessor. Let $\mathfrak{F}_x^{\text{alg}}$ be the finite set of these structural candidates. Physical common-point realization is not included in this definition; it is tested later by (206). Thus structural admissibility does not presuppose the physical conclusion.

The distinction is essential. The full pointed Hall/root–Bass model in [1] retains operation histories at the depth required by the fixed support, while a fixed-depth channel or face-minimal derived compression need not retain the fundamental-cycle chain (192). The prime-power all-support comparison preserves the structural cells once present, but it does not turn an arbitrary physical graph forest into an admissible one. Thus define

$$\kappa_{\text{mot}}(x) := \min_{F_x \in \mathfrak{F}_x^{\text{alg}}} \kappa_F(x),$$

with value $+\infty$ when $\mathfrak{F}_x^{\text{alg}} = \emptyset$. The unrestricted κ_{cyc} remains a valid purely scalar criterion; κ_{mot} is the corresponding criterion available to the motivic fundamental-cycle route. In particular,

$$\boxed{\text{ess sup}_x \kappa_{\text{mot}}(x) \leq \delta^{-o(1)}} \quad (193)$$

closes the scalar gate by the same forest-allocation proof as (191). If this condition fails but $\mathfrak{F}_x^{\text{alg}} \neq \emptyset$, the dense connected core of an admissible forest is the source passed to the hybrid comparison below.

10.11 Typed chord routing

There is nevertheless a genuine reduction in the typed comparison that must be proved. Write $b(e)$ for the unique root child whose descendant set contains $\gamma(e)$, and fix an order on the root children. Let F_x^{min} be the measurable minimizing forest chosen in (163), and let $E_x^{\text{ch,min}}$ be the once-oriented physical broad pairs (e, e') whose child edge $\{b(e), b(e')\}$ is a chord of F_x^{min} . For an admissible strict type ξ , write $\mathcal{P}_{\text{mot}}(e, e'; \xi)$ for the finite set of face/residue-support provenances in the existing pair complex which realize ξ from (e, e') . Let

$$\mathcal{P}_{\xi,x}^{\text{ch}} := \bigcup_{(e,e') \in E_x^{\text{ch,min}}} \mathcal{P}_{\text{mot}}(e, e'; \xi), \quad \mathcal{Y}_x^{\text{ch,lab}} := \bigsqcup_{\xi} \bigsqcup_{\rho \in \mathcal{P}_{\xi,x}^{\text{ch}}} \mathcal{Y}_{\xi,x}^{[2]}.$$

It is enough to construct a measurable joint lift on this chord source alone,

$$\Theta_x^{\text{ch}} : E_x^{\text{ch,min}} \longrightarrow \mathcal{Y}_x^{\text{ch,lab}},$$

which preserves x , uses an admissible strict type $\xi \in \mathcal{S}_{\text{alg}}(e, e')$, retains ρ until the branch trace, and satisfies $\rho \in \mathcal{P}_{\text{mot}}(e, e'; \xi)$ on each source pair. Let $\text{pr}_{\text{phys}}(\xi, \rho; h, h') := (\xi, h, h')$.

The required coalesced weighted-fibre estimate is

$$\boxed{\sum_{\substack{(e,e') \in E_x^{\text{ch,min}}: \\ \text{pr}_{\text{phys}} \Theta_x^{\text{ch}}(e,e') = (\xi, h, h')}} u_e(x) u_{e'}(x) \leq C_\delta v_{\xi,h}(x) v_{\xi,h'}(x), \quad C_\delta = \delta^{-o(1)}} \quad (194)$$

The projection in the sum forgets ρ only after all of its labelled contributions to the same physical destination have been coalesced. Summing (194), then using predecessor R6 and (159), gives

$$\begin{aligned}\mathfrak{C}_{I,k}^{\text{br}} &= 2 \int \sum_{(e,e') \in E_x^{\text{ch},\min}} u_e u_{e'} dx \\ &\leq 2C_\delta \sum_\xi \int \mathbf{N}_\xi^2 \lesssim \delta^{-o(1)} B_I^{\text{pred}} \int \mathbf{N}_{I,k}.\end{aligned}$$

Hence (194) proves (165) and closes the scalar root gate. No routing is needed on the forest edges for this implication. If a deterministic chord lift is too rigid, the same proof uses the measurable Hall cuts restricted to the chord-source measure.

The chord-to-predecessor target can be weakened once more by using the whole fundamental cycle. For this typed route choose measurably $F_x^{\text{mot}} \in \mathfrak{F}_x^{\text{alg}}$; if the admissible class is empty, this route has a structural obstruction before any estimate is attempted. With the fixed child order, set

$$\lambda_{I,k,x}^{\text{br}} := \sum_{\substack{b(e) < b(e'), \ T(e) \neq T(e') \\ e, e' \text{ broad at } x}} u_e(x) u_{e'}(x) \delta_{(e,e')}.$$

Let

$$E_x^{\text{ch},\text{mot}} := \{(e, e') \in E_x^{\text{br}} : \{b(e), b(e')\} \notin F_x^{\text{mot}}\}$$

be the chord source belonging to this selected admissible forest. It need not equal $E_x^{\text{ch},\min}$, which belongs to the unrestricted scalar minimizer. Disintegrate this once-oriented root measure into

$$\lambda_{I,k,x}^{\text{br}} = \lambda_{I,k,x}^F + \lambda_{I,k,x}^{\text{ch}},$$

according as the child edge lies in F_x^{mot} or its complement, and put

$$d\Lambda_{I,k}^F := 2 d\lambda_{I,k,x}^F dx, \quad d\Lambda_{I,k}^{\text{ch}} := 2 d\lambda_{I,k,x}^{\text{ch}} dx.$$

Then

$$\boxed{\|\Lambda_{I,k}^F\| \leq 2B_I^{\text{pred}} \int \mathbf{N}_{I,k}, \quad \|\Lambda_{I,k}^{\text{ch}}\| =: \mathfrak{C}_{I,k}^{\text{br}}(F^{\text{mot}}) \geq \mathfrak{C}_{I,k}^{\text{br}}.} \quad (195)$$

For a chord $\{b, b'\}$, let $P_{F_x^{\text{mot}}}(b, b')$ be its unique forest path. For a physical chord atom z , declare $z \rightsquigarrow_F y$ precisely when y occurs as a boundary face of a trace-compatible faithfully labelled motivic filling of its fundamental cycle, and y is either a compatible physical pair over an edge of $P_{F_x^{\text{mot}}}(b, b')$ or a strict typed predecessor. Thus

$$z \rightsquigarrow_F y, \quad y \in \mathcal{Z}_{I,k}^F \sqcup \bigsqcup_\xi \mathcal{Y}_\xi^{[2]},$$

where $\mathcal{Z}_{I,k}^F := \text{supp} \Lambda_{I,k}^F$, and compatibility includes the common physical point, source histories, orientation line and residue-support label ρ . If no such filling exists, the corresponding allowed neighbourhood is empty. For a measurable chord set E , write $\mathcal{A}_F(E)$ and $\mathcal{A}_\xi^{\text{ch}}(E)$ for its allowed neighbourhoods in the forest and predecessor targets. The exact hybrid capacity target is

$$\boxed{\Lambda_{I,k}^{\text{ch}}(E) \leq C_\delta \left[\Lambda_{I,k}^F(\mathcal{A}_F(E)) + \sum_\xi \Lambda_\xi^{\text{pred}}(\mathcal{A}_\xi^{\text{ch}}(E)) \right]} \quad (196)$$

for every measurable E , $C_\delta = \delta^{-o(1)}$.

As in (160), fibrewise max-flow/min-cut shows that this is equivalent to one measurable positive splitting kernel into the combined target, chosen once and then restricted hereditarily. Taking the whole chord source in (196), using (195), predecessor R6 and (159), yields

$$\mathfrak{C}_{I,k}^{\text{br}}(F^{\text{mot}}) \lesssim \delta^{-o(1)} B_I^{\text{pred}} \int \mathbf{N}_{I,k}.$$

Thus the hybrid Hall estimate also proves (165) and closes the root gate for the whole F^{mot} -decomposition. It is weaker than (194): forest boundary faces may absorb chord mass, but only up to their already bounded global capacity. A formal fundamental-cycle identity only names candidate faces. Constructing a trace-compatible physical filling is part of the realization interface, and even such a filling proves neither the Hall cuts nor a bound for their congestion.

10.12 Faithful positive cycle realization

Structural cycle closure and positive physical control must be kept separate. A coordinate lift may construct cycle classes and their Gersten boundaries, while triangular branch curvature survives compression; closure does not bound the faithfully labelled coefficient current. In the present dictionary, $\mathfrak{R}_{\text{alg}}$ is the structural cycle lift, whereas $\mathfrak{C}_{I,k}^{\text{br}}$ measures the positive physical mass left on its independent chords. The full motivic filtration names these typed cycles, but algebraic K -theory proves neither (165) nor Hall capacity. Its legitimate role is to organize the cycle-space obstruction whose positive physical mass must still be estimated.

10.13 A labelled chain model

More precisely, importing that interface here would require no new descent coordinate. The once-oriented measure introduced above satisfies

$$d\Lambda_{I,k}^{\text{root}} = 2 d\lambda_{I,k,x}^{\text{br}} dx, \quad J_x^{\text{br}} := (b(e), b(e'))_* \lambda_{I,k,x}^{\text{br}} = \sum_{b < b'} M_{bb'}(x) [b, b'].$$

Let $\tilde{C}_{\bullet}^{\text{mot}}$ be the subcomplex of the existing pair base change $\mathcal{K}^{[2]}$ generated by the typed root-collision cells and by the fundamental-cycle two-cells belonging to the selected motivic-admissible forest. The provenance set $\mathcal{P}_{\text{mot}}(e, e'; \xi)$ just introduced is a label inside the all-support filtration, not another filtration coordinate. Its physical root-collision current is the weighted degree-one chain

$$\mathfrak{C}_{I,k,x}^{\text{root}} := \sum_{\substack{b(e) < b(e'), \ T(e) \neq T(e') \\ e, e' \text{ broad at } x}} u_e(x) u_{e'}(x) [e, e']_{\text{mot}}.$$

For A in the finite physical pair fibre, write $\mathfrak{C}_{I,k,x}^{\text{root}}|_A$ and $\lambda_{I,k,x}^{\text{br}}|_A$ for the corresponding restrictions. These restrictions are taken inside one fixed realization; no new routing is chosen after A is specified.

Let $C_{\bullet}^{\text{str,lab}}(x)$ be the structural labelled chain complex whose one-generators are admissible quadruples

$$(e, e'; \xi, \rho), \quad \xi \in \mathcal{S}_{\text{alg}}(e, e'), \quad \rho \in \mathcal{P}_{\text{mot}}(e, e'; \xi).$$

Let $C_{\bullet}^{\text{lab}}(x)$ be its physical refinement whose one-generators are admissible sextuples

$$(e, e'; \xi, \rho; h, h'), \quad \xi \in \mathcal{S}_{\text{alg}}(e, e'), \quad \rho \in \mathcal{P}_{\text{mot}}(e, e'; \xi),$$

where $(h, h'; x) \in \mathcal{Y}_{\xi,x}^{[2]}$ is an allowed common-point physical lift of $(e, e'; x)$. Thus parallel strict-type labels, residue-support provenances, source histories and destination histories are not identified. Write

$$\tau_{\text{phys}} : C_{\bullet}^{\text{lab}}(x) \longrightarrow C_{\bullet}^{\text{str,lab}}(x)$$

for the map forgetting only (h, h') . Write π for the map forgetting all labels except the two root children, and let Tr_{br} forget $(\xi, \rho; h, h')$ but retain the physical source pair; thus $\pi = (b, b')_* \circ \text{Tr}_{\text{br}}$.

There are then exactly three structural requirements. First, the motivic pair-orientation line must lift to the physical branch-orientation line:

$$j_{e,e'}^{\text{mot}} : \mathfrak{o}_{\text{mot}}(e, e') \xrightarrow{\sim} \mathfrak{o}_{\text{br}}(e, e'), \quad j_{e,e'}^{\text{mot}} \text{ is branch-exchange equivariant.}$$

The faithfully labelled structural carrier can be constructed explicitly. Let $\mathcal{L}_x^{\text{str}}$ be the finite set of structural labelled cells in the selected Hall/root–Bass subcomplex and put

$$F_x^{\text{flag}} := \mathbf{Q}(z_\sigma : \sigma \in \mathcal{L}_x^{\text{str}}).$$

For every strict labelled flag $\sigma = (\sigma_0 < \dots < \sigma_j)$, set

$$\kappa_x^{\text{flag}}(\sigma) := (-1)^j \{z_{\sigma_0}, \dots, z_{\sigma_j}\} \in K_{j+1}^M(F_x^{\text{flag}}).$$

Writing $x_\sigma^0 = [z_\sigma = 0]$, its zero-face Gersten boundary is

$$\partial_0 \kappa_x^{\text{flag}}(\sigma) = \sum_{i=0}^j (-1)^i x_{\sigma_i}^0 \{z_{\sigma_0}, \dots, \widehat{z_{\sigma_i}}, \dots, z_{\sigma_j}\}. \quad (197)$$

Send each residue generator on the right to the labelled flag with σ_i deleted. Then

$$\mathcal{R}_x^{\text{flag}}(\partial_0 \kappa_x^{\text{flag}}(\sigma)) = \partial_{\text{str,lab}}[\sigma]. \quad (198)$$

The independent coordinates retain the deleted-cell label, so this realization is faithful before compression. In degree two it uses coordinate K_3 symbols to realize the fundamental-cycle fillings; coordinate K_4 symbols provide the coherence between alternative fillings. Consequently the structural labelled obstruction vanishes for every $F \in \mathfrak{F}_x^{\text{alg}}$. This is the legitimate algebraic K -theory contribution. It does not produce a lift through τ_{phys} , positivity, or a bound on physical coefficients.

After this orientation identification, one must construct, measurably in x , measure-valued linear maps

$$\nu_j^{\text{cyc}}(x) : \tilde{C}_j^{\text{mot}} \longrightarrow \mathcal{M}(C_j^{\text{lab}}(x))$$

where $\mathcal{M}$ denotes finite signed measures on the labelled generators, and satisfying

$$\begin{aligned} \partial_{\text{lab}} \circ \nu_2^{\text{cyc}} &= \nu_1^{\text{cyc}} \circ \partial_{\text{mot}}, \\ \nu_1^{\text{cyc}}(\mathbf{c}_{I,k,x}^{\text{root}}|_A) &\geq 0, \\ \text{Tr}_{\text{br}} \nu_1^{\text{cyc}}(\mathbf{c}_{I,k,x}^{\text{root}}|_A) &= \lambda_{I,k,x}^{\text{br}}|_A \quad \text{for every } A. \end{aligned} \quad (199)$$

The compressed identity

$$\pi_* \nu_1^{\text{cyc}}(\mathbf{c}_{I,k,x}^{\text{root}}|_A) = (b, b')_*(\lambda_{I,k,x}^{\text{br}}|_A)$$

then follows rather than being imposed separately.

The orientation lift, the first identity, and the last trace identity are, respectively, the double-orientation, zero matching-defect, and branch-mass-compatible trace conditions. A coordinate flag/ K -theory lift (197)–(198) constructs their structural labelled part. Retaining ρ , positivity of the realized degree-one current, and the branch-trace identity prevent compression from discarding a cycle detected by the physical branch trace. In particular, the map is not allowed to coalesce two equal ξ 's before their residue-support labels have been traced. These properties are not consequences of a signed Gersten lift. Even after (199) is proved, the quantitative estimate

(165) remains independent. Thus the cycle-space version of the full-motive route has two separate obligations: a faithful positive cycle realization and a weighted physical cycle budget, not merely formal vanishing of a motivic boundary. For the scalar root gate only the latter is needed; for the hereditary typed statement the exact capacity requirement remains (160).

For a structural candidate satisfying (192), the full Hall/root–Bass two-chain and the coordinate flag carrier already supply the faithfully provenance-labelled structural boundary. The remaining question is whether that boundary has a common-point positive physical lift with the prescribed branch trace and destination capacity. No separate obstruction layer is needed: all of these requirements enter the single finite system below. For a structural one-generator $g = (e, e'; \xi, \rho)$, define its common-point physical fibre

$$\mathcal{L}_x(g) := \{(h, h') : (e, e'; \xi, \rho; h, h') \in C_1^{\text{lab}}(x)\},$$

and put

$$\boxed{\mathcal{S}_{\text{phys}}(e, e'; x) := \{(\xi, \rho) : \xi \in \mathcal{S}_{\text{alg}}(e, e'), \rho \in \mathcal{P}_{\text{mot}}(e, e'; \xi), \mathcal{L}_x(e, e'; \xi, \rho) \neq \emptyset\}.} \quad (200)$$

Because the fibres are finite, nonemptiness of $\mathcal{S}_{\text{phys}}(e, e'; x)$ is exactly the measurable atomic support test. Existing insertion/deletion/confluence spans prove this test for the labels they realize, but support alone gives no pair RN bound.

10.14 External-product and correlated capacities

Let $\mathcal{S}_{\boxtimes}(e, e'; x) \subseteq \mathcal{S}_{\text{phys}}(e, e'; x)$ consist of labels realized by an endpointwise external product of two unary physical kernels, and put

$$E_x^{\boxtimes} := \{(e, e') \in E_x^{\text{br}} : \mathcal{S}_{\boxtimes}(e, e'; x) \neq \emptyset\}, \quad E_x^{\text{corr}} := E_x^{\text{br}} \setminus E_x^{\boxtimes}.$$

Fix one external-product label $\theta = (\xi, \rho, K_{\theta}^{(1)}, K_{\theta}^{(2)})$, and let $E_x(\theta) \subseteq E_x^{\boxtimes}$ be the pairs assigned to it by the fixed label order. Unary R5 gives

$$\sum_e u_e(x) K_{\theta}^{(1)}(h \mid e) \leq A_{\theta,1} v_{\xi,h}(x), \quad \sum_{e'} u_{e'}(x) K_{\theta}^{(2)}(h' \mid e') \leq A_{\theta,2} v_{\xi,h'}(x).$$

Nonnegativity and the external-product formula then give the pair estimate

$$\boxed{\begin{aligned} & \sum_{(e,e') \in E_x(\theta)} u_e u_{e'} K_{\theta}^{(1)}(h \mid e) K_{\theta}^{(2)}(h' \mid e') \\ & \leq \left(\sum_e u_e K_{\theta}^{(1)}(h \mid e) \right) \left(\sum_{e'} u_{e'} K_{\theta}^{(2)}(h' \mid e') \right) \leq A_{\theta,1} A_{\theta,2} v_{\xi,h} v_{\xi,h'}. \end{aligned}} \quad (201)$$

The tensor product of (14) on the two endpoints supplies the labelled matching identity, and its coefficients are positive with exact branch trace. Coalescing the finitely many θ -labels costs only the fixed operation-history alphabet and the subpower carrier factor already counted in (152). Hence (206) is proved on $E_x^{\boxtimes}$. The genuinely unresolved typed source is E_x^{corr} . The diagonal confluence example above shows why the same calculation cannot be extended to a correlated joint kernel.

10.15 Mode decomposition and fractional capacity

A concrete sufficient route on E_x^{corr} is to split it among coherent deterministic modes which retain $(e, e'; \rho)$, common-point support and the same motivic boundary. This is genuinely additional to

the merged comparison theorem: the Hall/root–Bass realization in [1] is faithful on structural operation words but does not assert local fullness or produce positive physical kernels.

Fix a labelled type $\theta = (\xi, \rho)$ and a coherent deterministic output

$$F_\theta(e, e') = (h_\theta(e, e'), h'_\theta(e, e'))$$

on every positive source pair. Its mode-conflict graph $\mathfrak{G}_{\theta, x}^{\text{mode}}$ has the source pairs as vertices; two vertices are adjacent when they share the first endpoint but have different first outputs, or share the second endpoint but have different second outputs.

A color class is exactly a source set on which the output separates as $(\phi(e), \psi(e'))$. Indeed, a convex combination equal to a point mass can use only product factors supported at that same point, so the active source of each factor is independent; conversely each color class defines the two endpoint maps. Consequently

$$\mathfrak{r}_+^{\text{mode}}(F_\theta; x) = \chi(\mathfrak{G}_{\theta, x}^{\text{mode}}). \quad (202)$$

For the diagonal confluence map $(i, j) \mapsto (i + j, i + j)$ on $(\mathbf{Z}/R)^2$, every color class is a matching and the R level sets $i + j = s$ are perfect matchings. Hence $\mathfrak{r}_+^{\text{mode}} = R$, although there is only one unlabelled structural operation type. Structural word count and the q^2 provenance therefore do not bound the physical mode number.

If several coherent outputs are allowed, minimize the right-hand side over the source-local choices before coalescing destination capacities. Removing $\beta_1(\mathfrak{G}_{\theta, x}^{\text{mode}})$ edges leaves a forest; choosing one endpoint of each removed edge gives a feedback vertex set of size at most β_1 . Color the remaining forest with two colors and give those vertices separate colors. Hence

$$\chi(\mathfrak{G}_{\theta, x}^{\text{mode}}) \leq 2 + \beta_1(\mathfrak{G}_{\theta, x}^{\text{mode}}). \quad (203)$$

Thus a subpower cycle-rank bound on the faithfully labelled mode-conflict graph is a valid sufficient condition for the positive mode gate. It is not the cycle rank of the compressed root-child interaction graph G_x^{br} , and the weighted destination capacities must still be coalesced as in (206).

Positive splitting permits a sharper weighted fractional form. Let $\mathcal{I}(\mathfrak{G}_{\theta, x}^{\text{mode}})$ be the independent source sets and write $a_C := A_{C,1}A_{C,2}$. Put

$$\begin{aligned} \kappa_+^{\text{mode}}(\theta, x) &:= \min \left\{ \sum_{C \in \mathcal{I}} a_C t_C : t_C \geq 0, \sum_{C \ni z} t_C \geq 1 \ \forall z \right\}, \\ \kappa_+^{\text{mode}}(\theta, x) &\leq \delta^{-o(1)} \implies (206) \text{ on this selector.} \end{aligned} \quad (204)$$

For a feasible family, set

$$\alpha_C(z) := \frac{t_C \mathbf{1}_C(z)}{\sum_{C' \ni z} t_{C'}}.$$

Then $\sum_C \alpha_C(z) = 1$, $\alpha_C(z) \leq t_C$, and the output separates endpointwise on C . Applying (201) on every C proves the implication in (204). The exact weighted finite LP dual is

$$\kappa_+^{\text{mode}}(\theta, x) = \max \left\{ \sum_z \omega_z : \omega_z \geq 0, \sum_{z \in C} \omega_z \leq a_C \text{ for every independent } C \right\}. \quad (205)$$

A large dual value therefore extracts one coalesced weighted physical source on which no coherent separated mode carries more than its unary RN cost. This is the correct positive mode-core

obstruction; estimating independently selected local color classes would again be a local-to-global error. An integral coloring gives $\kappa_+^{\text{mode}} \leq \chi(\mathfrak{G}_{\theta,x}^{\text{mode}}) \max_C a_C$, so (203) remains a stronger sufficient shadow.

Within this deterministic-mode route, the remaining sufficient statement is

$$\text{ess sup}_x \sum_{\theta} \kappa_+^{\text{mode}}(\theta, x) \leq \delta^{-o(1)}.$$

This is the physical no-new-communication bound for the joint first-obstruction selector, with provenance and common-point labels retained. It cannot be replaced by formal termination, operation-word faithfulness, or the q^2 line.

10.16 Finite linear programming formulation

Now let $\nu_{\bullet}^{\text{str}}$ be a chosen coordinate-flag structural chain map. Keep the source atom as part of every index. Order all available degree-one physical labels over the positive source atoms, put their coefficients in a_x , and put the available source-labelled signed higher-cell coefficients in b_x . Missing physical support simply removes the corresponding column and makes the system below infeasible if that structural coefficient is required. The boundary, fixed structural pushforward and source-by-source branch-trace equations have the form $M_x a_x + N_x b_x = c_x$. No source labels are coalesced, so deleting source blocks implements every later restriction without a new choice. Let $D_x a_x$ be the vector obtained by coalescing all degree-one labelled coefficients at each physical predecessor destination, and let q_x be the corresponding vector of capacities $v_{\xi,h}(x)v_{\xi,h'}(x)$; for the hybrid route append the forest reservoir coordinates. The single typed realization target is

$$\boxed{\begin{array}{ll} \text{find } a_x \geq 0, & b_x \text{ signed, such that} \\ M_x a_x + N_x b_x = c_x, & D_x a_x \leq C_{\delta} q_x, \quad C_{\delta} = \delta^{-o(1)}. \end{array}} \quad (206)$$

The first equation simultaneously imposes the fixed structural pushforward, the labelled chain identity and the source-by-source branch trace. Hence a solution of (206) is already a source-local coherent positive physical lift with the required RN bound. Conversely every such lift gives these coefficients.

There is also one exact dual test. Introduce a nonnegative slack vector in the capacity inequality. Finite-dimensional Farkas duality says that (206) is feasible exactly when

$$\boxed{\langle y, c_x \rangle + C_{\delta} \langle \lambda, q_x \rangle \geq 0 \quad \text{whenever} \quad N_x^{\top} y = 0, \quad M_x^{\top} y + D_x^{\top} \lambda \geq 0, \quad \lambda \geq 0.} \quad (207)$$

Because all source labels remain uncoalesced, the lexicographically first feasible basic solution is measurable in x , and later restriction deletes source blocks from that same solution. If a coherent filling has already been fixed on every allowed source–destination edge, the variables b_x can be eliminated and (207) reduces to the ordinary Hall cuts (160). In general it is the correct replacement: support (200) and Hall capacity are merely projections; graded matching is encoded by the linear rows and positivity by $a_x \geq 0$.

Taking $A = E_x^{\text{ch}, \text{mot}}$ in (199) supplies the structural labelled chord lift required above, possibly as a positive splitting kernel rather than a deterministic Θ_x^{ch} ; the matching identity for the two-cells supplies its trace-compatible candidate boundary faces. What it does not supply is the hybrid destination-capacity estimate (196); the direct chord estimate (194) is a stronger sufficient alternative. Thus restriction compatibility connects the full motivic interface to the scalar cycle reduction without turning a collection of locally reselected maps into a global one.

For the selected F_x^{mot} , if its congestion exceeds a proposed level K , let $H_x \subseteq F_x^{\text{mot}}$ be the minimal connected witness obtained from (190) with $F^{\circ} = F_x^{\text{mot}}$, and restrict the physical source to

$$E_x^{\text{core}}(H_x) := \{(e, e') \in E_x^{\text{ch}, \text{mot}} : P_{F_x^{\text{mot}}}(b(e), b(e')) \subseteq H_x\}.$$

Applying the same source-local realization to this single restriction retains the whole bridgeless cycle core and disintegrates it by the existing $(\xi, \rho; h, h')$ labels. This is the correct input to the full all-support descent: the first failed typed obstruction is taken on the core current, not independently on its chord atoms. No conclusion follows merely from the existence of those labels. Different cores can have overlapping predecessor neighbourhoods, so their capacities must remain coalesced in the single Hall ledger; summing separately proved core estimates would reintroduce the forbidden local-to-global step.

In particular, $\beta_1(G_x^{\text{br}})$ is not a Hall theorem. It counts excess scalar interactions in the compressed root-child graph. Cycles in the faithfully labelled routing multigraph instead describe possible redistributions among admissible destinations; without target capacities they give neither a positive routing nor its Radon–Nikodym bound.

Thus (158), rather than another fixed-root Córdoba estimate, is the natural full-motive target of the Selmer–Cartan route. The scalar gate could of course be proved by a different physical argument; the pair realization is the form compatible with the descent filtration. The broad packet selection has already been normalized in the source-mass ledger; it does not by itself prove the pair-measure domination in (158).

10.17 The algebraic candidate and the remaining physical gate

The root-cross quantity does have a canonical algebraic controller. The divided-power identity

$$\gamma_2(y + z) - \gamma_2(y) - \gamma_2(z) = yz$$

gives, after the real realization $\gamma_2(t) = t^2/2$,

$$\mathbf{P}_{I,k}^{\text{root}}(x) := \gamma_2(W_{r_*}(x)) - \sum_{b \in \text{ch}(r_*)} \gamma_2(W_b(x)), \quad (208)$$

and the history partition yields

$$\boxed{\mathbf{P}_{I,k}^{\text{root}} = \sum_{b < b'} W_b W_{b'} \geq 0, \quad \mathbf{X}_{I,k}^{\text{root}} = 2 \int \mathbf{P}_{I,k}^{\text{root}}.} \quad (209)$$

The factor $1/2$ is used only over the real physical leg; the integral controller requires no division by 2. The positivity in (209) appears only after the three PD terms have been combined. The term $\gamma_2(y + z)$ depends jointly on both inputs, while the two unary terms are subtracted. Hence the polarization identity is not a termwise positive decomposition and cannot itself define the kernels $\mathcal{R}_\xi$, prove the joint weighted-fibre estimate (161), or verify the Hall cuts.

This is intrinsically a second-moment object. Its physical source is the pair diagonal

$$\Omega_{I,k}^{[2]} := \bigsqcup_{\gamma, T, \gamma', T'} \{(x; \gamma, T; \gamma', T') : u_{\gamma, T}(x) u_{\gamma', T'}(x) > 0\}, \quad d\mu_{I,k}^{[2]} := u_{\gamma, T}(x) u_{\gamma', T'}(x) dx, \quad (210)$$

the fibre product of two marked history sources over the same physical point. Accordingly the relevant realization is the external pair base change

$$\mathcal{K}^{[2]} := \mathcal{K} \boxtimes_{\mathbf{R}^d} \mathcal{K} \quad (211)$$

over the fixed (η, ν) -fibre, with the existing support, confluence and Cartan filtrations on both factors. The kernels in (158) must positively route its broad root-collision part by the first failed typed obstruction. This is not another motivic level: it is the physical realization forced by the quadratic estimate R6.

Two firewalls are important. First, both factors have the same operation support, so support-squarefree local faithfulness does not apply; the unrestricted PD controller records

the repeated input but supplies no positive measure partition. Second, the physical product $u_\gamma u_{\gamma'}$ is polarization in $\Gamma^2(D_I)$, whereas $L_{q^2}^{\text{conf}} = W_q/q^2 W_q$ records the supplied primitive filtered confluence provenance. The product neither creates that lift nor changes coefficient depth. The merged prime-power all-support comparison already handles the latter, and algebraic K -theory is not an analytic input here.

For a root-collision pair the PD hull nevertheless gives a finite algebraic candidate. Put

$$\text{Pol}(\gamma, \gamma') := \gamma_2(h_\gamma + h_{\gamma'}) - \gamma_2(h_\gamma) - \gamma_2(h_{\gamma'}),$$

apply the support-functorial filtration, and take its first nonzero typed graded component; if it vanishes, use the finite Reedy reconstruction. The result is

$$\mathfrak{R}_{\text{alg}}(\gamma, \gamma') \in \bigoplus_{\xi \prec (\eta; I, \chi, \nu, n)} \mathbf{Z}[\xi]. \quad (212)$$

This chain lowers (213), but it may have signs. Therefore the routing kernels must be an additional positive measurable refinement of $\mathfrak{R}_{\text{alg}}$, with the RN ledger (158). Formal Reedy termination does not imply that inequality. Concretely, the missing refinement is a positive joint physical lift whose weighted fibres satisfy (161), or a splittable lift satisfying (160). Signed PD cancellation proves neither.

11 Where the two-dimensional theorem enters

For a rooted d -packet, the full direction support has cardinality d : one root slot and $d - 1$ side slots. The unified tower's exact-support comparison is used for $|I| \geq 3$. Its last root projection reaches $|I| = 2$, the ordinary Kummer/Heisenberg low sector, containing one root and one side direction. The physical realization of this terminal two-dimensional piece is the hereditary planar Córdoba estimate [2]

$$\int_{\mathbf{R}^2} m_Z^2 \lesssim N\delta(1 + \log(1/\delta)), \quad m_Z \lesssim_\zeta \delta^{-\zeta} \frac{1 + \log(1/\delta)}{\lambda}$$

after deletion of an $O(\delta^\zeta)$ fraction of shaded incidence mass. With the slowly varying choice of ζ made after (113), this is the base case for (124). In particular, when $\lambda_2 \geq \delta^{o(1)}$, it gives $B_{K_2} \lesssim \delta^{-o(1)}$. The proper-predecessor recombination above reduces R6 to (142). Conditional on that estimate, ordinary finite induction on $|I|$, using the latching, confluence/load and fresh exact-support terms at each stage, propagates the estimate through the support filtration. The forgotten root coordinate is restored by (113) before the next extension step. This is induction in the physical realization of the already constructed typed coefficient filtration, not an intrinsic construction of motives.

12 Finite descent

Fix the finite support, decoration, history, and Cartan ceilings from the standing scope. If $N_s = \prod_{p \in \Pi} p^{a_p}$, set

$$\mathfrak{c}(\eta; I, \chi, \nu, n; N_s) := |\eta|_{\text{val}} + |I| + |\nu|_1 + \ell(\chi) + n + \sum_{p \in \Pi} a_p. \quad (213)$$

A proper support face lowers $|I|$, a strict predecessor lowers $\ell(\chi)$, a decoration reduction lowers its own depth, and a Cartan predecessor lowers n . Exact-order loss replaces N_s by a proper divisor. The structural interchange identities ensure that lowering one coordinate does not increase another, so every branch is finite.

At each state the audit (30) determines the next proper piece. On the physical leg, a zero determinant is reconstructed from proper faces; a nonzero determinant passes to the root antecedent and then to one lower support. The identity (3) records the parallel motivic normalization but is not used to infer physical nonvanishing. Passive decoration fibres disappear under (31); history/load fibres remain in (2).

Consequently formal termination carries no analytic gain. Conditional on (142), the closed support recurrence and the two-dimensional base estimate control every support stage. The root-cross gate is indispensable: neither the complexity decrease nor motivic reconstruction proves a global incidence estimate.

13 Closure of the filtered route

13.1 Structural bookkeeping

The fixed-root determinant comparison (8), literal side faces, root projection, insertion kernels, and the symmetric packet law (17) verify the finite physical filtration. Global gluing and every hereditary restriction are controlled by (24) and (25). The operators (27) give the structural representation (28); the passive decoration base change (1) does not alter physical multiplicity.

After the augmentation (31), the only analytic data are the exact-support determinant, the confluence/load fibre, and the root antecedent. The reduced-history tree and (137) control every strict predecessor. Thus the global confluence estimate R6 is equivalent to the remaining physical root-cross gate (142); no formal motivic reconstruction is used to prove it.

13.2 Scalar analytic closure

In dimension two the hereditary planar Córdoba estimate [2] supplies the base case. In dimension three, (52)–(59), together with the sticky inputs (177), close the root gate.

At the three-to-four-dimensional handoff, the factorized and fibre-disjoint blocks close by (108) and (109). Directionwise sampling and the unbiased lift close every shell covered by (91). Proper-history fibre collisions are absorbed by (73); the residual low-occupancy source is exactly (93). Its broad part is multilinear, its narrow part retains the affine fibre mark, and the uniform marked theorem [9] supplies (81). Hence the scalar route closes through dimension four.

For $d \geq 5$, the analogous source-hereditary marked root gate remains an input. Assuming (142) at every retained support and load stage, the support induction (125) gives

$$B_{K_I} \lesssim_d \delta^{-o(1)}.$$

The final weighted-to-unweighted cutoff (130) then retains subpower incidence mass. For sparse initial shading the same argument keeps the explicit density factors in (124); the filtration creates no density.

13.3 The separate typed requirement

The hereditary typed statement requires a positive common-point routing on the correlated pair source. Its exact forms are the Hall/Radon–Nikodym condition (160) and the equivalent capacity system (207); the fractional-mode bound (204) is sufficient. The structural tower supplies labelled destinations and boundaries, but neither this positivity nor the weighted capacity. Consequently the scalar theorem above does not claim the typed comparison.

13.4 Terminal incidence count

Finally convert the incidence set $E \subset \Omega_{\text{inc}}$ into the terminal ordinary descendant

$$Y'(T) := \{x \in Y(T) : (Q(x), T, x) \in E\}, \quad m'(x) := \sum_T \mathbf{1}_{Y'(T)}(x) = m_E(x).$$

The incidence identity gives

$$\sum_T |Y'(T)| = \int m'(x) dx \leq \|m'\|_\infty \left| \bigcup_T Y'(T) \right|. \quad (214)$$

If $\mathcal{I}_{\text{in}} := \sum_T |Y(T)|$, then $\mathcal{I}_{\text{in}} = W$ and the cell partition gives

$$\sum_T |Y'(T)| = W \mu_{\text{inc}}(E).$$

Thus (130) gives

$$\sum_T |Y'(T)| \geq \frac{\alpha}{2} \mathcal{I}_{\text{in}}.$$

Under the same root-cross hypothesis, the mass ledger and terminal cutoff therefore imply

$$\left| \bigcup_T Y'(T) \right| \gtrsim_d \delta^{o(1)} \mathcal{I}_{\text{in}}.$$

In particular, for N_T direction-separated tubes with $|Y(T)| \geq \lambda_{\text{in}} \delta^{d-1}$,

$$\left| \bigcup_T Y'(T) \right| \gtrsim_d \delta^{o(1)} N_T \lambda_{\text{in}} \delta^{d-1}.$$

This last passage uses only incidence counting. After taking the scalar root gate as a physical hypothesis, the unproved analytic content is confined to (142), rather than hidden in a local-to-global root assembly. The already realized q^2 confluence provenance is structural data and is not used as an analytic estimate in this implication.

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